

THE STRAIGHT LINE



SYNOPSIS

Inclination of a line:

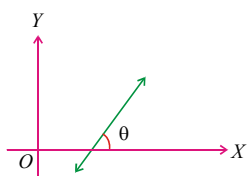
- If a line makes an angle $\theta (0 \leq \theta < \pi)$ with x-axis measured in positive direction then θ is called inclination of the line.

i) Inclination of horizontal line is zero

ii) Inclination of vertical line is $\pi/2$

Slope of a line:

- If the inclination of a non vertical line is θ then $\tan \theta$ is called slope of the line and is usually denoted by m , thus $m = \tan \theta$



- i) Slope of horizontal line (x-axis) is zero

$$(\because \theta = 0^\circ)$$

- ii) Slope of vertical line (y-axis) is not defined

$$(\because \theta = 90^\circ)$$

- iii) $\theta = 0^\circ \Leftrightarrow m = 0$; $0^\circ < \theta < 90^\circ \Leftrightarrow m > 0$

$$\theta = 90^\circ \Leftrightarrow m \text{ is not defined}$$

$$90^\circ < \theta < 180^\circ \Leftrightarrow m < 0$$

- Slope of the line joining two points $A(x_1, y_1)$,

$$B(x_2, y_2) \text{ is } m = \frac{y_2 - y_1}{x_2 - x_1} \quad (x_1 \neq x_2)$$

- i) If $x_1 = x_2$ then the line $\overline{AB}$ is vertical and hence its slope is not defined

- ii) If $y_1 = y_2$ then the line $\overline{AB}$ is horizontal and hence its slope is 0

- Two nonvertical lines are parallel if their slopes are equal.

- Two non vertical lines are perpendicular if product of their slopes is -1

W.E-1: The medians AD and BE of the triangle with vertices $A(0, b)$, $B(0, 0)$ and $C(a, 0)$ are mutually perpendicular if

Sol: $AD \perp BE \Rightarrow \left(\frac{-2b}{a}\right)\left(\frac{b}{a}\right) = -1; \Rightarrow 2b^2 = a^2$

- If θ is an angle between two nonvertical lines having slopes m_1, m_2 then

$$\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1 m_2}, m_1 m_2 \neq -1$$

i) If θ is acute then $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

- ii) If θ is one angle between two lines then the other angle is $\pi - \theta$. Usually the acute angle between two lines is taken as the angle between the lines

Intercept(s) of a line:

- If a line cuts x-axis at $A(a, 0)$ and y-axis at $B(0, b)$ then a and b are called x-intercept and y-intercept of that line respectively

- i) Intercept of a line may be positive or negative or zero

- ii) x-intercept of a horizontal line is not defined

- iii) y-intercept of a vertical line is not defined

- iv) Intercepts of a line passing through origin are zero.

Equation of a straight line in various forms:

- **i) Line parallel to x-axis:** Equation of horizontal line passing through (a, b) is $y = a$

- ii) Line parallel to y-axis:** Equation of vertical line passing through (a, b) is $x = b$

- iii) Slope - point form :** The equation of the line with slope m and passing through the point (x_1, y_1) is $y - y_1 = m(x - x_1)$

W.E-2: If $(3, -1), (2, 4), (-5, 7)$ are the mid points of the sides $\overline{BC}, \overline{CA}, \overline{AB}$ of triangle ABC .

Then the equation of the side $\overline{CA}$ is

Sol : Here $m = -1$ and given point (x_1, y_1) is $(2, 4)$.
By point slope form equation of the line is
 $y - 4 = -1(x - 2)$

- iv) Two - point form :** The equation of a line passing through two points

$A(x_1, y_1)$ and $B(x_2, y_2)$ is

$$(y - y_1)(x_2 - x_1) = (x - x_1)(y_2 - y_1)$$

$$\text{(or)} \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

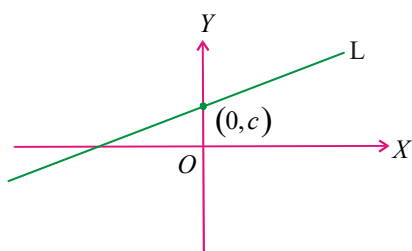
W.E-3: Equation of the diagonal (through the origin) of the quadrilateral formed by the lines $x = 0$, $y = 0$, $x + y = 1$ and $6x + y = 3$ is

Sol : Here $(x_1, y_1) = (0, 0)$, $(x_2, y_2) = \left(\frac{2}{5}, \frac{3}{5}\right)$

Using two-point form, the equation of the line is $3x - 2y = 0$

v) Slope - Intercept form :

a) The equation of the line whose slope is m and which cuts an intercept 'c' on the y-axis is $y = mx + c$



b) The equation of the line whose slope is m and which cuts an intercept 'a' on the x-axis is $y = m(x - a)$

c) The equation of the line passing through the origin and having slope m is $y = mx$

W.E-4: Equation to the straight line cutting off an intercept 2 from negative y axis and inclined at 30° to the positive direction of axis of x, is

Sol : Equation of line passing through $(0, -2)$ and having slope $\frac{1}{\sqrt{3}}$ is $\sqrt{3}y - x + 2\sqrt{3} = 0$

vi) Intercept Form : Suppose a line L makes intercept on x-axis is a and on y-axis is b then its equation is $\frac{x}{a} + \frac{y}{b} = 1$

a) If the portion of the line intercepted between the axes is divided by the point (x_1, y_1) in the ratio $m : n$, then the equation of the line is $\frac{nx}{x_1} + \frac{my}{y_1} = m + n$

(or) $\frac{mx}{x_1} + \frac{ny}{y_1} = m + n$

b) Equation of the line whose intercept between the axes is bisected at the point (x_1, y_1) is $\frac{x}{x_1} + \frac{y}{y_1} = 2$

c) Equation of the line making equal intercepts on the axes and through the point (x_0, y_0) is

$$x + y = x_0 + y_0$$

d) Equation of the line making equal intercepts in magnitude but opposite in sign and passing through (x_0, y_0) is $x - y = x_0 - y_0$

e) The equation of the line passing through the point (x_1, y_1) and whose intercepts are in the ratio $m : n$ is $nx + my = nx_1 + my_1$ (or) $mx + ny = mx_1 + ny_1$

W.E-5: The sum of x,y intercepts made by the lines $x + y = a$, $x + y = ar$, $x + y = ar^2$ on coordinate axes when $r = 1/2$, $a \neq 0$

Sol : required sum

$$= 2a + 2ar + 2ar^2 + \dots (\text{infinite G.P})$$

$$= 2a / (1 - r) = 4a$$

vii) General equation of line :

a) A linear equation in x and y always represents a line.

b) The equation of a line in general form is $ax + by + c = 0$, where a, b, c are real numbers such that $a^2 + b^2 \neq 0$ having slope $= -a/b$,

x-intercept $= -c/a$, y-intercept $= -c/b$.

c) The equation of a line parallel to $ax + by + c = 0$ is of the form $ax + by + k = 0, k \in R$.

d) The equation of a line perpendicular to $ax + by + c = 0$ is of the form $bx - ay + k = 0, k \in R$

e) Equation of a line passing through (x_1, y_1) and (i) parallel to $ax + by + c = 0$ is $a(x - x_1) + b(y - y_1) = 0$

(ii) Perpendicular to $ax + by + c = 0$ is $b(x - x_1) - a(y - y_1) = 0$

viii) Normal form :

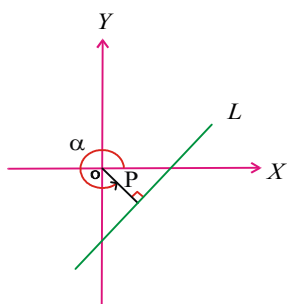
a) The equation of the straight line upon which the length of the normal drawn from origin is 'p' and this perpendicular makes an angle α , $(0 \leq \alpha < 2\pi)$ with positive x-axis is $x \cos \alpha + y \sin \alpha = p, (p > 0)$

b) The equation of the straight line upon which the length of the normal drawn from origin is 'p' and this perpendicular makes an angle α , $(0 \leq \alpha < 2\pi)$ with positive x-axis is $x \cos \alpha + y \sin \alpha = p, (p > 0)$

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b) The normal form of a line $ax + by + c = 0$ is

$$\frac{(-a)}{\sqrt{a^2+b^2}}x + \frac{(-b)}{\sqrt{a^2+b^2}}y = \frac{c}{\sqrt{a^2+b^2}}, \text{ if } c > 0$$

and

$$\frac{a}{\sqrt{a^2+b^2}}x + \frac{b}{\sqrt{a^2+b^2}}y = \frac{-c}{\sqrt{a^2+b^2}}, \text{ if } c < 0$$

W.E-6: Normal form of the equation $x+y+1=0$ is

Sol: The given equation is $x+y+1=0 \Rightarrow -x-y=1$

$$\Rightarrow \frac{(-1)x}{\sqrt{2}} + \frac{(-1)y}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x \cos\left(\pi + \frac{\pi}{4}\right) + y \sin\left(\pi + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow x \cos \frac{5\pi}{4} + y \sin \frac{5\pi}{4} = \frac{1}{\sqrt{2}}$$

ix) Symmetric form and Parametric equations of a straight line :

a) The equation of the straight line passing through (x_1, y_1) and makes an angle θ with the positive

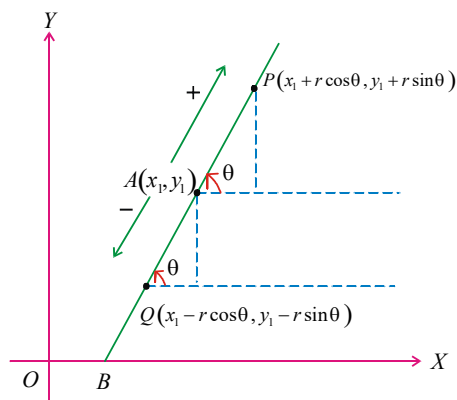
direction of x-axis is $\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta}$

Where $\theta \in (0, \pi/2) \cup (\pi/2, \pi)$

b) The co-ordinates (x, y) of any point P on the line at a distance 'r' units away from the point $A(x_1, y_1)$ can be taken as

$(x_1 + r \cos \theta, y_1 + r \sin \theta)$ (or) $(x_1 - r \cos \theta, y_1 - r \sin \theta)$

c) The equations $x = x_1 + r \cos \theta$, $y = y_1 + r \sin \theta$ are called parametric equations of a line with parameter 'r' of the line passing through the point (x_1, y_1) and having inclination θ .



$$\cos \theta = \frac{x - x_1}{AP}, \quad \sin \theta = \frac{y - y_1}{AP}$$

$$\text{or } x - x_1 = AP \cos \theta, \quad y - y_1 = AP \sin \theta.$$

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$$

W.E-7: $(1,2), (3,6)$ are two opposite vertices of a rectangle and if the other two vertices lie on the line $2y = x + c$, then c and other two vertices are

Sol: Mid point of given vertices is

$P(x_1, y_1) = (2, 4)$ which lies on $2y = x + c$ then

$$c = 6. \text{ Now } r = BP = AP = \sqrt{5}, \quad \tan \theta = \frac{1}{2}$$

$$\text{Hence } B = (x_1 + r \cos \theta, y_1 + r \sin \theta) = (4, 5)$$

$$C = (x_1 - r \cos \theta, y_1 - r \sin \theta) = (0, 3)$$

Distances:

➤ i) The perpendicular distance to the line $ax + by + c = 0$

$$(a) \text{ from origin is } \frac{|c|}{\sqrt{a^2 + b^2}}$$

$$(b) \text{ from the point } (x_1, y_1) \text{ is } \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

ii) The distance of a point (x_1, y_1) from the line $L \equiv ax + by + c = 0$ measured along a line making an

$$\text{angle } \alpha \text{ with x-axis is } \left| \frac{ax_1 + by_1 + c}{a \cos \alpha + b \sin \alpha} \right|$$

iii) The distance between parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is

$$\frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

iv) The distance between the parallel lines $ax+by+c_1=0$ and $ax+by+c_2=0$ measured along the line having inclination θ is

$$\left| \frac{c_1 - c_2}{a \cos \theta + b \sin \theta} \right|$$

v) The equation of a line parallel and lying midway between the above two lines is

$$ax + by + \frac{c_1 + c_2}{2} = 0$$

vi) Equation of the line parallel to $ax+by+c=0$ and at a distance d from the line is

$$ax + by + c \pm d\sqrt{a^2 + b^2} = 0$$

W.E-8: The distance between $A(2, 3)$ on the line of gradient $3/4$ and the point of intersection P of this line with $5x + 7y + 40 = 0$ is

Sol : Since $m = 3/4$, then $\cos \theta = 4/5$ and $\sin \theta = 3/5$.

$$r = \frac{\left| 5 \times 2 + 7 \times 3 + 40 \right|}{\left| 5 \left(\frac{4}{5} \right) + 7 \left(\frac{3}{5} \right) \right|} = \frac{355}{41}$$

Position of a point (s) w.r.to line (s):

➤ i) The ratio in which the line $L \equiv ax + by + c = 0$ divides the line segment joining

$A(x_1, y_1)$ and $B(x_2, y_2)$ is $-L_{11} : L_{22}$ where

$$L_{11} = ax_1 + by_1 + c, L_{22} = ax_2 + by_2 + c$$

ii) The points A, B lie on the same side or opposite side of the line $L = 0$ according as L_{11}, L_{22} have same sign or opposite sign that is $L_{11} \cdot L_{22} > 0$ or $L_{11} \cdot L_{22} < 0$

W.E-9: The range of θ in the interval $(0, \pi)$ such that the points $(3, 5)$ and $(\sin \theta, \cos \theta)$ lie on the same side of the line $x + y - 1 = 0$ is

Sol : Since $(3 + 5 - 1)(\sin \theta + \cos \theta - 1) > 0$

$$\Rightarrow \sin \left(\frac{\pi}{4} + \theta \right) > \frac{1}{\sqrt{2}} \Rightarrow \frac{\pi}{4} < \frac{\pi}{4} + \theta < \frac{3\pi}{4}$$

$$\Rightarrow 0 < \theta < \frac{\pi}{2}$$

iii) A point $A(x_1, y_1)$ and origin lies on the same or opposite side of a line $L = ax + by + c = 0$

according as $c \cdot L_{11} > 0$ or $c \cdot L_{11} < 0$

iv) The point (x_1, y_1) lies between the parallel lines $ax_1 + by_1 + c = 0$, $ax_2 + by_2 + c = 0$ or does

not lie between them according as $\frac{ax_1 + by_1 + c_1}{ax_1 + by_1 + c_2}$ is negative or positive

v) The point $A(x_1, y_1)$ lies above or below the line $L = ax + by + c = 0$ according as

$$\frac{L_{11}}{b} > 0 \text{ or } \frac{L_{11}}{b} < 0$$

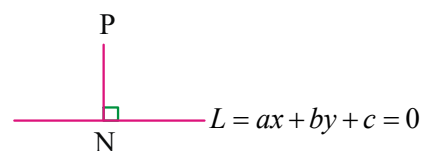
Proof: The fig. Shows a point $P(x_1, y_1)$ lying above a given line. If an ordinate is dropped from P to meet the line L at N , then the x coordinate of N will be x_1 . Putting $x = x_1$ in the equation $ax + by + c = 0$ gives

$$\text{ordinate of } N = -\frac{(ax_1 + c)}{b}$$

If $P(x_1, y_1)$ lies above the line, then we have

$$y_1 > -\frac{(ax_1 + c)}{b} \quad \text{i.e.} \quad y_1 + \frac{(ax_1 + c)}{b} > 0$$

$$\text{i.e.} \quad \frac{(ax_1 + by_1 + c)}{b} > 0, \quad \text{i.e.} \quad \frac{L(x_1, y_1)}{b} > 0$$



Hence, $P(x_1, y_1)$ lies above the line

$ax + by + c = 0$, and if $\frac{L(x_1, y_1)}{b} < 0$, it would mean that P lies below the line $ax + by + c = 0$.

Ceva's Theorem :

➤ If the lines joining any point 'O' to the vertices A, B, C of a triangle meet the opposite sides in $D, E,$

$$F \text{ respectively then } \frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$$

Proof: Without loss of generality take the point P as the origin O .

Let $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$ be the

vertices. Slope of $\overline{AP}$ is $\frac{y_1 - 0}{x_1 - 0} = \frac{y_1}{x_1}$

Equation of $\overline{AP}$ is $y - 0 = \frac{y_1}{x_1}(x - 0)$

$$\Rightarrow yx_1 = xy_1 \Rightarrow xy_1 - yx_1 = 0$$

$$\therefore \frac{BD}{DC} = \frac{-(x_2y_1 - x_1y_2)}{x_3y_1 - x_1y_3} = \frac{x_1y_2 - x_2y_1}{x_3y_1 - x_1y_3}$$

$$\therefore \frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$$

Manelaw's Theorem :

- If a transversal cuts the sides BC, CA, AB of a triangle in D, E, F respectively then

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1$$

Proof: Let the transversal be $ax+by+c=0$. the line

divides $\overline{BC}$ at D then $\frac{BD}{DC} = \frac{-(ax_2+by_2+c)}{(ax_3+by_3+c)}$

$$\text{Hence } \frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1$$

W.E-10: The range of α , if (α, α^2) lies inside the triangle having sides along the lines

$$2x + 3y = 1, x + 2y - 3 = 0, 6y = 5x - 1$$

Sol : Let A, B, C be vertices of the triangle.

$$A = (-7, 5), B = \left(\frac{5}{4}, \frac{7}{8}\right)$$

$$C = \left(\frac{1}{3}, \frac{1}{9}\right). \text{ Sign of A w.r.t. BC to -ve.}$$

If P lies inside the triangle ABC, then sign of P will be the same as sign of A w.r.t. the line BC

$$\Rightarrow 5\alpha - 6\alpha^2 - 1 < 0 \dots (i)$$

$$\text{similarly } 2\alpha + 3\alpha^2 - 1 > 0 \dots (ii)$$

$$\text{And } \alpha + 2\alpha^2 - 3 < 0 \dots (iii)$$

Solving (i), (ii) and (iii) for α and then taking intersection,

$$\text{we get } \alpha \in \left(\frac{1}{2}, 1\right) \cup \left(-\frac{3}{2}, -1\right)$$

Point of intersection of lines and Concurrency of Straight Lines:

- i) Consider two lines $L_1 \equiv a_1x + b_1y + c_1 = 0$ and $L_2 \equiv a_2x + b_2y + c_2 = 0$ then point of intersection is

$$\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}\right) \text{ or}$$

$$\frac{x}{\begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}} = \frac{y}{\begin{vmatrix} c_1 & a_1 \\ c_2 & a_2 \end{vmatrix}} = \frac{1}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

ii) Three or more lines are said to be concurrent, if they have a point in common. The common point is called the point of concurrence.

a) If $L_1 = 0, L_2 = 0$ are two intersecting lines, then the equation of any line other than

$L_1 = 0$ and $L_2 = 0$ passing through point of intersection can be taken as

$L_1 + \lambda L_2 = 0$. Where λ is a parameter

b) The three lines $L_i \equiv a_ix + b_iy + c_i = 0, i = 1, 2, 3$ are

$$\text{concurrent iff } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

(or) Point of intersection of any two lines lies on the third line

(or) there exist constants $\lambda_1, \lambda_2, \lambda_3$ not all zero such that $\lambda_1 L_1 + \lambda_2 L_2 + \lambda_3 L_3 = 0$

c) If $p_1x + q_1y = 1, p_2x + q_2y = 1, p_3x + q_3y = 1$ are concurrent lines then the points $(p_1, q_1), (p_2, q_2), (p_3, q_3)$ are collinear

d) If $ka + lb + mc = 0$, then the point of concurrency of the lines represented by $ax + by + c = 0$ is

$$\left(\frac{k}{m}, \frac{l}{m}\right)$$

W.E-11: The line $x + \lambda y - 4 = 0$ passes through the point of intersection of $4x - y + 1 = 0$ and $x + y + 1 = 0$. Then the value of λ is

Sol : The three lines are concurrent

$$\Rightarrow \begin{vmatrix} 1 & \lambda & -4 \\ 4 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow -2 - 3\lambda - 20 = 0 \Rightarrow \lambda = -\frac{22}{3}$$

Angle between lines:

- i) If ' θ ' is an acute angle between the lines having

$$\text{slopes } m_1 \text{ and } m_2 \text{ then } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

ii) If ' θ ' is an acute angle between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ then

$$\cos\theta = \left| \frac{a_1a_2 + b_1b_2}{\sqrt{a_1^2 + b_1^2} \sqrt{a_2^2 + b_2^2}} \right| \text{ and } \tan\theta = \left| \frac{a_1b_2 - a_2b_1}{a_1a_2 + b_1b_2} \right|$$

other angle between the lines is $\pi - \theta$

iii) The slope m of a line which is equally inclined with two intersecting lines of slopes m_1 and m_2

$$\text{is given by } \frac{m_1 - m}{1 + mm_1} = \frac{m - m_2}{1 + mm_2}$$

iv) The slopes of the lines making an angle α with

$$\text{a line having slope } m \text{ are } \frac{m - \tan\alpha}{1 + m \tan\alpha}, \frac{m + \tan\alpha}{1 - m \tan\alpha}$$

v) Consider two lines $L_1 = a_1x + b_1y + c_1 = 0$

and $L_2 = a_2x + b_2y + c_2 = 0$

a) Lines are parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$

b) Lines are coincident if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

c) Lines are perpendicular if $a_1a_2 + b_1b_2 = 0$

d) Lines are equally inclined with x-axis

$$\text{if } \frac{a_1}{a_2} = -\frac{b_1}{b_2}$$

W.E-12: A straight line through (2, 2) intersects the lines $\sqrt{3}x + y = 0$ and $\sqrt{3}x - y = 0$ at the points A and B. The equation to the line AB so that the triangle OAB is equilateral is

Sol: Since given two lines passing through origin and making angles $60^\circ, 120^\circ$ with X-axis the third line is parallel to X-axis. Hence equation of AB is $y=2$

Triangles and Quadrilaterals:

➤ i) The ratio of the sides of a triangle formed by $L_1 = 0, L_2 = 0$ and $L_3 = 0$ is

$$\sqrt{a_1^2 + b_1^2} \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} : \sqrt{a_2^2 + b_2^2} \begin{vmatrix} a_3 & b_3 \\ a_1 & b_1 \end{vmatrix} : \sqrt{a_3^2 + b_3^2} \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

ii) Let d_1 be the distance between the

parallel lines $ax + by + c_1 = 0, ax + by + c_2 = 0$

and d_2 be the distance between the parallel lines

$a_1x + b_1y + k_1 = 0, a_1x + b_1y + k_2 = 0$ then the figure formed by four lines is

a) a square if $d_1 = d_2$ and $aa_1 + bb_1 = 0$,

b) Rhombus if $d_1 = d_2$ and $aa_1 + bb_1 \neq 0$,

c) Rectangle if $d_1 \neq d_2$ and $aa_1 + bb_1 = 0$,

d) Parallelogram if $d_1 \neq d_2$ and $aa_1 + bb_1 \neq 0$

➤ i) The area of triangle formed by the line

$$\frac{x}{a} + \frac{y}{b} = 1 \text{ with the co-ordinate axis is } \frac{1}{2} |ab|$$

ii) The area of triangle formed by line $ax + by + c = 0$ with the co-ordinate axes is

$$\frac{c^2}{2|ab|}$$

iii) Area of the rhombus $a|x| + b|y| + c = 0$ is

$$4(\text{area of } \Delta) = \frac{2c^2}{|ab|}$$

iv) The area of triangle formed by lines

$$a_ix + b_iy + c_i = 0, i = 1, 2, 3 \text{ is } \frac{\Delta^2}{2|\lambda_1\lambda_2\lambda_3|}$$

$$\text{where } \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, \lambda_1 = \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix},$$

$$\lambda_2 = \begin{vmatrix} a_1 & b_1 \\ a_3 & b_3 \end{vmatrix}, \lambda_3 = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

v) The area of triangle formed by lines

$$y = m_ix + c_i, i = 1, 2, 3 \text{ is } \frac{1}{2} \left| \sum \frac{(c_1 - c_2)^2}{m_1 - m_2} \right|$$

vi) If p_1, p_2 are distances between parallel sides and ' θ ' is angle between adjacent sides of

parallelogram then its area is $\left| \frac{p_1 p_2}{\sin\theta} \right|$

vii) Area of parallelogram whose sides $a_1x + b_1y + c_1 = 0, a_1x + b_1y + c_2 = 0, a_2x + b_2y + d_1 = 0$ and $a_2x + b_2y + d_2 = 0$ is

$$\left| \frac{(c_1 - c_2)(d_1 - d_2)}{a_1b_2 - a_2b_1} \right|$$

viii) Area of rhombus = $\frac{1}{2} d_1 d_2$ where d_1, d_2 are lengths of the diagonals

W.E-13: The triangle formed by the lines $x-7y-22=0$, $3x+4y+9=0$, $7x+y-54=0$ is

Sol: by using

$$\sqrt{a_1^2+b_1^2} \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} : \sqrt{a_2^2+b_2^2} \begin{vmatrix} a_3 & b_3 \\ a_1 & b_1 \end{vmatrix} : \sqrt{a_3^2+b_3^2} \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

we get $1 : \sqrt{2} : 1$ hence the triangle is right angled isosceles.

W.E-14: If the distance of any point $P(x, y)$ from the origin is defined as $d(x, y) = \text{Max. } (|x|, |y|)$ and $d(x, y) = a$ (non zero constant), then the locus of the P is

Sol: $d(x, y) = \text{Max. } (|x|, |y|)$ (i)

But $d(x, y) = a$ (ii)

From (i) and (ii),

$$a = \text{Max. } (|x|, |y|)$$

If $|x| > |y|$, then $a = |x| \Rightarrow x = \pm a$

If $|y| > |x|$, then $a = |y| \Rightarrow y = \pm a$

Hence locus of P represents a square.

Foot and Image:

➤ i) If (h, k) is the foot of the perpendicular from (x_1, y_1) to the line $ax + by + c = 0$ then

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = \frac{-(ax_1+by_1+c)}{a^2+b^2} \text{ or}$$

$$(h, k) = (x_1 + a\lambda, y_1 + b\lambda) \text{ where}$$

$$\lambda = \frac{-(ax_1+by_1+c)}{a^2+b^2}$$

ii) If (h, k) is the image (reflection) of the point

(x_1, y_1) w.r.t the line $ax + by + c = 0$ then

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = \frac{-2(ax_1+by_1+c)}{a^2+b^2} \text{ or}$$

$$(h, k) = (x_1 + a\lambda, y_1 + b\lambda) \text{ where}$$

$$\lambda = \frac{-2(ax_1+by_1+c)}{a^2+b^2}$$

iii) Image of (a, b) w.r.to $y = x$ is (b, a)

iv) Image of (a, b) w.r.to $x + y = 0$ is $(-b, -a)$

v) If B is image of A w.r.to P then $2P = A + B$

vi) Reflection of $f(x, y) = 0$ in x -axis is

$$f(x, -y) = 0$$

vii) Reflection of $f(x, y) = 0$ in y -axis is

$$f(-x, y) = 0$$

viii) Reflection of $f(x, y) = 0$ in $x = y$ is

$$f(y, x) = 0$$

ix) Image of the line $ax + by + c = 0$ w.r.t line $lx + my + n = 0$ (or) the straight line $lx + my + n = 0$ bisects an angle between the two lines of which one of them is $ax + by + c = 0$ then equation of other line is

$$(l^2 + m^2)(ax + by + c) = 2(al + bm)(lx + my + n)$$

W.E-15: In $\triangle ABC$ A is $(1, 2)$ if the internal angle bisector of B is $2x - y + 10 = 0$ and perpendicular bisector of AC is $y = x$ then the equation of BC is

Sol: Image of A w.r.to bisector of B is $(-7, 6)$ lies on BC and image of A in the perpendicular bisector of AC is $C(2, 1)$

$\therefore$ equation of BC is $5x + 9y - 19 = 0$

Centroid, circumcentre, orthocentre and incentre:

➤ i) Let $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$ be vertices of $\triangle ABC$ then,

a) Equation of altitude through A is

$$(x - x_1)(x_2 - x_3) = (y - y_1)(y_2 - y_3)$$

b) Equation of perpendicular bisector of the side $\overline{AB}$ is

c) Orthocentre of $\triangle ABC$ is

$$\left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C} \right)$$

d) Circumcentre of $\triangle ABC$ is

$$\left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C} \right)$$

ii) The equations of the sides BC, CA and AB of the triangle ABC formed by the lines

$$L_i = a_i x + b_i y + c_i = 0 \quad (i=1, 2, 3) \text{ then}$$

- a) Orthocentre is point of intersection of
 $(a_2a_3 + b_2b_3)L_1 = (a_3a_1 + b_3b_1)L_2 = (a_1a_2 + b_1b_2)L_3$
- b) Median through A is $\lambda_2L_2 - \lambda_3L_3 = 0$
- Hence centroid satisfies $\lambda_2L_2 = \lambda_3L_3 = \lambda_1L_1$
- iii) If H is orthocentre of triangle ABC, then orthocentre of triangle formed by any three of the points H, A, B, C will be the remaining point.
- iv) Circumcentre is equidistant from the vertices of triangle
- v) If G is the centroid, H is the orthocentre and S is the circumcentre then
- a) The relation between them is $3G = 2S + H$.
- b) $H = 3G$ when $S = (0, 0)$
- vi) Incentre is equidistant from all sides of the triangle.
- vii) In a triangle ABC,
- a) The internal bisector of angle A, ie. AD divides opposite side BC at D in the ratio AB:AC
- b) The external bisector of angle A, ie. AD divides opposite side BC at D in the ratio -AB:AC
- viii) If the algebraic sum of the perpendicular distances from three points to a variable line is zero, then the line passes through the centroid of the triangle formed by the three points.

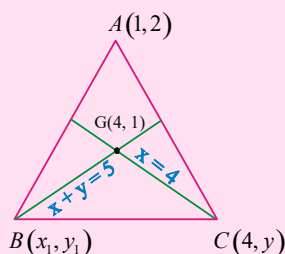
W.E-16: In a triangle ABC, coordinates of A are (1, 2) and the equations to the medians through B and C are $x + y = 5$ and $x = 4$ respectively. Then the points B and C are

Sol: Let B be (x_1, y_1) and C be $(4, y)$. Since medians through B and C meet at Centroid G (4, 1)

$$\Rightarrow \frac{x_1 + 4 + 1}{3} = 4 \Rightarrow x_1 = 7$$

Since $B(x_1, y_1)$ lies on $x + y = 5$

$$\therefore y_1 = 5 - x_1 = 5 - 7 = -2 \Rightarrow B \text{ is } (7, -2),$$



$$\text{Also } \frac{y_1 + y + 2}{3} = 1 \Rightarrow y = 3 - 2 - y_1$$

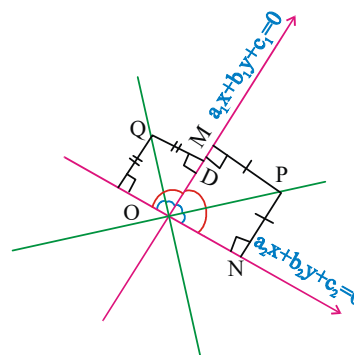
$$\therefore C \text{ is } (4, 3)$$

Angular bisectors of two straight lines:

- Angular bisector is the locus of a point which moves in such a way so that its distance from two intersecting lines remains same.

The equations of the two bisectors of the angles between the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$



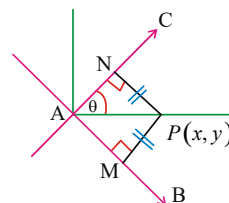
- i) If the two given lines are not perpendicular i.e. $a_1a_2 + b_1b_2 \neq 0$ and not parallel i.e. $a_1b_2 \neq a_2b_1$ then one of these equations is the equation of the bisector of the acute angle between two given lines and the other that of the obtuse angle between two given lines.

- ii) Whether both given lines are perpendicular or not, but the angular bisectors of these lines will always be mutually perpendicular.

iii) The bisectors of the acute and the obtuse angles:

Take one of the lines and let its slope be m_1 and take one of the bisectors and let its slope be m_2 . If θ be the acute angle between them, then find $\tan \theta$

$$= \left| \frac{m_1 - m_2}{1 + m_1m_2} \right|$$



If $\tan \theta > 1$ then the bisector taken is the bisector of the obtuse angle and the other one will be the bisector of the acute angle.

If $0 < \tan \theta < 1$ then the bisector taken is the bisector of the acute angle and the other one will be the bisector of the obtuse angles.

iv) consider the lines are $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, where $c_1 > 0$, $c_2 > 0$ then,

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

will represent the equation of the bisector of the acute or obtuse angle between the lines according as $a_1a_2 + b_1b_2$ is negative or positive.

v) **The equation of the bisector of the angle which contains a given point :**

The equation of the bisector of the angle between the two lines containing the point (x_1, y_1) is

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

$$\text{or } \frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = - \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

according as $a_1x_1 + b_1y_1 + c_1$ and $a_2x_1 + b_2y_1 + c_2$ are of the same signs or of opposite signs.

vi) For example the equation of the bisector of the angle containing the origin is given by

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}} \text{ for same sign}$$

of c_1 and c_2 (for opposite sign take -ve sign in place of +ve sign)

vii) If $c_1c_2(a_1a_2 + b_1b_2) < 0$, then the origin will lie in the acute angle and if $c_1c_2(a_1a_2 + b_1b_2) > 0$, then origin will lie in the obtuse angle.

viii) Equation of straight lines passing through $P(x_1, y_1)$ and equally inclined with the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are those which are parallel to the bisectors between these two lines and passing through the point P.

W.E-17 :

For the straight lines $4x + 3y - 6 = 0$ and $5x + 12y + 9 = 0$, find the equation of the -

- Bisector of the obtuse angle between them is
- Bisector of the acute angle between them is
- Bisector of the angle which contains origin is
- Bisector of the angle which contains (1, 2) is

Sol: after making $c_1 > 0$ and $c_2 > 0$;

$$a_1a_2 + b_1b_2 = (-4)(5) + (-3)(12) = -56 < 0$$

i) The bisector of the acute angle is

$$\frac{-4x - 3y + 6}{\sqrt{(-4)^2 + (-3)^2}} = \frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}}$$

$$7x + 9y - 3 = 0$$

ii) The bisector of the obtuse angle is

$$\frac{-4x - 3y + 6}{\sqrt{(-4)^2 + (-3)^2}} = - \frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}}$$

$$9x - 7y - 41 = 0$$

(iii) The bisector of the angle containing the origin

$$\frac{-4x - 3y + 6}{\sqrt{(-4)^2 + (-3)^2}} = \frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}} ; 7x + 9y - 3 = 0$$

(iv) For the point (1, 2),

$$4x + 3y - 6 = 4 \times 1 + 3 \times 2 - 6 > 0$$

$$5x + 12y + 9 = 12 \times 2 + 9 > 0$$

Hence equation of the bisector of the angle containing the point (1, 2) is

$$\frac{4x + 3y - 6}{5} = \frac{5x + 12y + 9}{13} ; 9x - 7y - 41 = 0$$

Optimization:

➤ Let A and B are two points on same side of line

$$L \equiv ax + by + c = 0$$

i) The point P such that PA + PB is minimum, is intersection of $L = 0$ and the line joining A to image of B

or line joining B to image of A w.r.to $L = 0$

ii) The point is P such that $|PA - PB|$ is

Maximum, is point of intersection of line $L = 0$ and line joining A and B.

W.E-18: A light ray emerging from the point source placed at $P(2, 3)$ is reflected at a point 'Q' on the y-axis and then passes through the point $R(5, 10)$. Coordinate of 'Q' is -

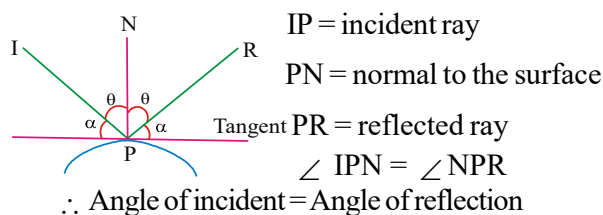
Sol: Image of point $P(2, 3)$ in Y-axis is $P^1(-2, 3)$

$$\text{Equation of } P^1R \equiv y - 3 = 1(x + 2)$$

$$x - y + 5 = 0$$

P^1R meets the Y-axis at $Q(0, 5)$

Reflection in surface:



No. of lines, no. of triangles and no. of circles:

- No. of lines drawn through the point A which are at a distance d from the point B
- If $AB = d$ then the no. of lines through A at a distance d from B is 1
 - If $AB > d$ then the no. of lines through A at a distance d from B is 2
 - If $AB < d$ then the no. of lines through A at a distance d from B is 0
- No. of right angled triangles in a circle depends on height h of the triangle and radius r of the circle
- If $h = r$, no. of right angled triangles = 2
 - If $h < r$, no. of right angled triangles = 4
 - If $h > r$, no. of right angled triangles = 0
- No. of circles touching three lines
- No circle if the lines are parallel
 - one circle if the lines are concurrent
 - 2 circles if two lines are parallel and third cuts them
 - 4 circles if the lines are not concurrent and no two of them are parallel.

W.E-19: Let $A = (1, 2)$, $B = (3, 4)$ and $C = (x, y)$ be a point such that $(x-1)(x-3) + (y-2)(y-4) = 0$. If area of $\triangle ABC = 1$ then maximum number of positions of C in the xy plane is

Sol: Ends of diameter are $A = (1, 2)$, $B = (3, 4)$

Area of the triangle is equal to $1 \Rightarrow \frac{1}{2}(2\sqrt{2})(h) = 1$

$$\Rightarrow h = \frac{1}{\sqrt{2}}$$

$$\text{radius} = \frac{AB}{2} = \frac{\sqrt{5}}{2}$$

$\therefore$ number of triangles = 4 ($\because h < r$)

**C.U.Q**

1. The straight line through $A(a, b)$ intersects the line through $B(c, d)$ at 'P' at right angles. The locus of P is

- $(x-a)(x-c) + (y-b)(y-d) = 0$
- $(x-a)(x-c) - (y-b)(y-d) = 0$
- $(x-b)(x-d) + (y-a)(y-d) = 0$
- $(x-b)(x-d) + (y-a)(y-c) = 0$

2. If $ax + by + c = 0$ is parallel to x-axis then which of the following is defined

- $\frac{a^2 - c^2}{c^2 + b^2}$
- $\frac{b^2 - c^2}{a^2}$
- $\frac{4b^2 + c^2}{\sqrt{abc}}$
- $\frac{\sqrt{a + c^2}}{a}$

3. The straightline $ax + by + c = 0$ ($a, b, c \neq 0$) will pass through the first quadrant and cut the positive x-axis, if

- $ac > 0, bc > 0$
- $ac > 0, bc < 0$
- $ac > 0$ and / or $bc > 0$
- $ac < 0$ and / or $bc < 0$

4. If $\frac{ax}{\sqrt{a^2 + b^2}} + \frac{by}{\sqrt{a^2 + b^2}} = \frac{-c}{\sqrt{a^2 + b^2}}$ is perpendicular form of a straight line then

- $a, b, c > 0$
- $a > 0, b > 0, c < 0$
- $c < 0$
- $a, b > 0, c \neq 0$

5. The straight line passing through $P(x_1, y_1)$ and making an angle α with x-axis intersects $Ax + By + C = 0$ in Q then $PQ =$

$$1) \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}} \quad 2) \frac{|Ax_1 + By_1 + C|}{\sqrt{A \cos \alpha + B \sin \alpha}}$$

$$3) \frac{|Ax_1 + By_1 + C|}{|A \cos \alpha + B \sin \alpha|} \quad 4) \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 \cos^2 \alpha + B^2 \sin^2 \alpha}}$$

6. If $a + b + c \neq 0$, $ax + by + c = 0$, $bx + cy + a = 0$, $cx + ay + b = 0$ are concurrent then

$$\frac{a^2 + b^2 + c^2}{ab + bc + ca} =$$

- 1/2
- 2
- 1
- 0

7. The lines

$$(a+b-2c)x + (b+c-2a)y + (c+a-2b) = 0,$$

$$(b+c-2a)x + (c+a-2b)y + (a+b-2c) = 0 \text{ and}$$

$$(c+a-2b)x + (a+b-2c)y + (b+c-2a) = 0 \text{ where } a, b, c, \text{ real numbers}$$

1) Form an equilateral triangle

2) Concurrent

3) Form a right angled triangle

4) Form an isosceles triangle

8. If the lines

$$p_1x + q_1y = 1, p_2x + q_2y = 1 \text{ and } p_3x + q_3y = 1$$

be concurrent, then the points

$$(p_1, q_1), (p_2, q_2) \text{ and } (p_3, q_3),$$

1) are collinear

2) form an equilateral triangle

3) form a scalene triangle

4) form a right angled triangle

9. $m \equiv a_1x + b_1y + c_1 = 0$ and $l \equiv a_2x + b_2y + c_2 = 0$ are

two straight lines such that $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ then

$m + kl = 0, k \in R$ is

- 1) a straight line different from m and l
- 2) not a straight line
- 3) is a straight line concurrent with m and l
- 4) the same straight line $m = 0$

10. If a and b are the intercepts made by the straight line on the coordinate axes such that

$\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$ then the line passes through point

- 1) $(1,1)$
- 2) (c,c)
- 3) $\left(\frac{1}{c}, \frac{1}{c}\right)$
- 4) $\left(\frac{c}{a}, \frac{c}{a}\right)$

11. The orthocentre of the triangle formed by the points $A(a \cos \alpha, a \sin \alpha)$

$B(a \cos \beta, a \sin \beta)$ $C(a \cos \gamma, a \sin \gamma)$ is

- 1) $(\cos \alpha + \cos \beta + \cos \gamma, \sin \alpha + \sin \beta + \sin \gamma)$
- 2) $[a(\cos \alpha + \cos \beta + \cos \gamma), a(\sin \alpha + \sin \beta + \sin \gamma)]$
- 3) $[a(\cos \alpha + \sin \beta + \sin \gamma), a(\sin \alpha + \cos \beta + \cos \gamma)]$
- 4) $(\cos \alpha \cos \beta \cos \gamma, \sin \alpha \sin \beta \sin \gamma)$

12. $(a, b), (c, d), (e, f)$ are the vertices of an equilateral triangle. Then the orthocentre of the triangle is

- 1) $\left(\frac{a+d+f}{3}, \frac{b+c+e}{3}\right)$
- 2) $\left(\frac{a+c+e}{3}, \frac{b+d+f}{3}\right)$
- 3) $\left(\frac{a+c+f}{3}, \frac{b+d+e}{3}\right)$
- 4) $\left(\frac{a+b+c}{3}, \frac{d+e+f}{3}\right)$

13. A triangle is formed by the lines $ax+by+c=0$ $lx+my+n=0$, $px+qy+r=0$, then the straight line $(ax+by+c)(lp+mq)=(lx+my+n)(ap+bq)$ passes through of the triangle.

- 1) Incentre
- 2) Circumcentre
- 3) Orthocentre
- 4) Centroid

14. The area of the triangle formed by the lines $y = m_1x + c_1$, $y = m_2x + c_2$ and $x=0$ is

- 1) $\frac{|c_1 - c_2|}{2|m_1 - m_2|}$
- 2) $\frac{(c_1 - c_2)^2}{2(m_2 - m_1)^2}$
- 3) $\frac{(c_1 - c_2)^2}{2|m_1 - m_2|}$
- 4) $\frac{(c_1 - c_2)^2}{|m_2 - m_1|}$

C.U.Q - KEY

- 1) 1
- 2) 1
- 3) 4
- 4) 3
- 5) 3
- 6) 3
- 7) 2
- 8) 1
- 9) 4
- 10) 2
- 11) 2
- 12) 2
- 13) 3
- 14) 3

C.U.Q - HINTS

1. Locus of P is a circle with end points of the diameter A,B
2. Line parallel to x-axis; $\Rightarrow$ x-coefficient = 0
3. $-\frac{c}{a} > 0$
4. $x \cos \alpha + y \sin \alpha = p$, where $p > 0$
5. let $PQ = r$
 $Q(x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$ sub in $Ax + By + C = 0$
6. $a^3 + b^3 + c^3 - 3abc = 0$
 $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 0$
7. $L_1 + L_2 + L_3 = 0$
8. Area = 0
9. $m = 0, l = 0$ represent coincident lines
10. $\frac{c}{a} + \frac{c}{b} = 1$
11. If $S = 0$ then $H = 3G$
12. $S = G$
13. Given equation is altitude
14. Use area of triangle formula



LEVEL - I (C.W)

SLOPE OF A LINE

1. If the line passing through the points $(-8,3)$ $(2,1)$ is parallel to the line passing through the points $(11,-1)$ $(k,0)$ then the value of k is
1) 5 2) 7 3) $5/2$ 4) 6
2. If each of the points $(a,4), (-2,b)$ lies on the line joining the points $(2,-1), (5,-3)$, then the point (a,b) lies on line
1) $6x + 6y - 25 = 0$ 2) $x + 3y + 1 = 0$
3) $2x + 6y + 1 = 0$ 4) $2x + 3y - 5 = 0$

3. If the lines $y = -3x + 4$, $ay = x + 10$ and $2y + bx + 9 = 0$ represent three consecutive sides of a rectangle then $ab =$

1) 18 2) -3 3) $\frac{1}{2}$ 4) $-\frac{1}{3}$

SLOPE-INTERCEPT FORM, SLOPE-POINT FORM AND TWO POINT FORM

4. The equation of the straight line cutting off an intercept 8 on x -axis and making an angle of 60° with the positive direction of y -axis is
- 1) $x - \sqrt{3}y - 8 = 0$ 2) $x + \sqrt{3}y = 8$
 3) $y - \sqrt{3}x = 8$ 4) $y + \sqrt{3}x = 8$
5. If $(-4, 5)$ is one vertex and $7x - y + 8 = 0$ is one diagonal of a square, then the equation of the other diagonal is
- 1) $x + 7y - 31 = 0$ 2) $x + 7y - 15 = 0$
 3) $x + 7y + 8 = 0$ 4) $x + 7y - 35 = 0$

INTERCEPTS AND INTERCEPT FORM

6. Equation of a line which passes through the point $(-3, 8)$ and cut off positive intercepts on the axes whose sum is 7 is
- 1) $3x - 4y = 12$ 2) $4x + 3y = 12$
 3) $3x + 4y = 12$ 4) $4x - 3y = 12$
7. The number of lines that are parallel to $2x + 6y - 7 = 0$ and have an intercept 10 between the co-ordinate axes is
- 1) 1 2) 2 3) 4 4) infinitely many
8. If the line $(x - y + 1) + k(y - 2x + 4) = 0$ makes equal intercept on the axes then the value of k is
- 1) $1/3$ 2) $3/4$ 3) $1/2$ 4) $2/3$

NORMAL FORM AND SYMMETRIC FORM

9. The equation of set of lines which are at a constant distance 2 units from the origin is
- 1) $x + y + 2 = 0$ 2) $x + y + 4 = 0$
 3) $x \cos \alpha + y \sin \alpha = 2$
 4) $x \cos \alpha + y \sin \alpha = \frac{1}{2}$
10. The slope of a straight line through $A(3, 2)$ is $3/4$ then the coordinates of the two points on the line that are 5 units away from A are
- 1) $(-7, 5)$ $(1, -1)$ 2) $(7, 5)$ $(-1, -1)$
 3) $(6, 9)$ $(-2, 4)$ 4) $(7, 3)$ $(-2, 1)$

PROBLEMS ON DISTANCES

11. The length of the perpendicular from the point $(0, 0)$ to the straight line passing through $P(1, 2)$ such that P bisects the intercepted part between axes
- 1) $\sqrt{5}$ 2) 4 3) $4/\sqrt{5}$ 4) $\sqrt{5}/4$
12. Radius of the circle touching the lines $3x + 4y - 14 = 0$, $6x + 8y + 7 = 0$ is (EAM-2011)
- 1) 7 2) $\frac{7}{2}$ 3) $\frac{7}{4}$ 4) $\frac{7}{6}$
13. The distance between the parallel lines given by $(x + 7y)^2 + 4\sqrt{2}(x + 7y) - 42 = 0$ is (EAM-2012)
- 1) 1 2) 5 3) 6 4) 2
14. Equation of the straight line parallel to $x + 2y - 5 = 0$ and at the same distance from $(3, 2)$ is
- 1) $x + 2y - 8 = 0$ 2) $x + 2y + 9 = 0$
 3) $x + 2y - 9 = 0$ 4) $x + 2y - 7 = 0$
15. If the straight line drawn through the point

$P(\sqrt{3}, 2)$ making an angle $\frac{\pi}{6}$ with x -axis

meets the line $\sqrt{3}x - 4y + 8 = 0$ at Q . Then PQ is

1) 4 2) 5 3) 6 4) 9

POSITION OF A POINT (S) W.R.T. LINE (S)

16. If the line $3x + 4y - 8 = 0$ is denoted by L , then the points $(2, -5), (-5, 2)$
- 1) lie on L 2) lie on same side of L
 3) lie on opposite sides of L
 4) equidistant from L
17. The vertices of a triangle are $O(0, 0)$, $B(-3, -1)$, $C(-1, -3)$. The equation of the line parallel to BC and intersecting the sides OB and OC whose perpendicular distance from O is $1/2$ is
- 1) $x + y = 1/\sqrt{2}$ 2) $x + y = -1/\sqrt{2}$
 3) $x + y = -1/2$ 4) $x + y = 1/2$
- ### POINT OF INTERSECTION OF LINES AND CONCURRENCY OF LINES
18. If the lines $ax + by + c = 0$, $bx + cy + a = 0$ and $cx + ay + b = 0$ $a \neq b \neq c$ are concurrent then the point of concurrency is
- 1) $(0, 0)$ 2) $(1, 1)$ 3) $(2, 2)$ 4) $(-1, -1)$

19. If the lines $3x+2y-5=0$, $2x-5y+3=0$, $5x+by+c=0$ are concurrent then $b+c =$
 1) 7 2) -5 3) 6 4) 9
20. The lines $(p-q)x + (q-r)y + (r-p) = 0$
 $(q-r)x + (r-p)y + (p-q) = 0$
 $(r-p)x + (p-q)y + (q-r) = 0$
 1) Form an equilateral triangle
 2) Form an Isosceles triangle
 3) are Concurrent
 4) Form a right angled triangle
21. If 2 is a root of $ax^2+bx+c=0$ then point of concurrence of lines $ax+2by+3c=0$ is
 1) (12,3) 2) (4,2) 3) (1,2) 4) (2, 3)
22. For all values of 'a' the set of straight lines $(3a+1)x - (2a+3)y + 9-a=0$ passes through the point
 1) (3, 4) 2) (4,2) 3) (3,3) 4) (1,2)
23. Equation of the line passing through the point of intersection of the lines $2x+3y-1=0$, $3x+4y-6=0$ and perpendicular to $5x-2y-7=0$ is (EAM- 2009)
 1) $2x+5y-19=0$ 2) $2x+5y+17=0$
 3) $2x+5y-16=0$ 4) $2x+5y-22=0$

ANGLE BETWEEN LINES

24. If $2x+3y+4=0$ & $\lambda x+ky+2=0$ are identical lines then $3\lambda-2k =$
 1) 1 2) 0 3) -1 4) 2
25. The angle between the diagonals of a quadrilateral formed by the lines $\frac{x}{a} + \frac{y}{b} = 1$,

$$\frac{x}{b} + \frac{y}{a} = 1, \frac{x}{a} + \frac{y}{b} = 2, \frac{x}{b} + \frac{y}{a} = 2 \text{ is}$$

- 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

TRIANGLES AND AREA OF THE TRIANGLE

26. The triangle formed by the lines $\sqrt{3}x+y-2=0$, $\sqrt{3}x-y+1=0$, $y=0$ is
 1) Equilateral 2) Right angled
 3) Right angled isosceles 4) Isosceles
27. The area of the triangle formed by the lines $x=0$; $y=0$ and $x \sin 18^\circ + y \cos 36^\circ + 1 = 0$ is
 1) 1 2) 2 3) 3 4) 4
28. If a straight line perpendicular to $3x-4y-6=0$ forms a triangle with the coordinate axes whose area is 6sq. units, then the equation of the straight line (s) is (EAM- 2014)

- 1) $x-2y=6$ 2) $4x+3y=12$
 3) $4x+3y+24=0$ 4) $3x-4y=12$
29. The equation of base of an equilateral triangle is $x+y=2$ and the vertex is (2, -1). Then area of triangle is
 1) $2\sqrt{3}$ 2) $\sqrt{3}/6$ 3) $1\sqrt{3}$ 4) $2\sqrt{3}$

QUADRILATERALS AND AREA OF THE QUADRILATERALS

30. The quadrilateral formed by the lines $2x-5y+7=0$, $5x+2y-1=0$, $2x-5y+2=0$, $5x+2y+3=0$ is
 1) Rectangle 2) Square
 3) Parallelogram 4) Rhombus
31. The diagonals of a parallelogram PQRS are along the lines $x+3y=4$ and $6x-2y=7$. Then PQRS must be :
 1) rectangle 2) square
 3) cyclic quadrilateral 4) rhombus

FOOT AND IMAGE

32. Foot of the perpendicular of origin on the line joining the points $(a \cos \theta, a \sin \theta)$, $(a \cos \phi, a \sin \phi)$ is
 1) $(\cos \theta + \cos \phi, \sin \theta + \sin \phi)$
 2) $(\cos \theta - \cos \phi, \sin \theta - \sin \phi)$
 3) $\left(\frac{a(\cos \theta + \cos \phi)}{2}, \frac{a(\sin \theta + \sin \phi)}{2} \right)$
 4) $(\cos \theta \cos \phi, \sin \theta \sin \phi)$
33. Suppose A, B are two points on $2x-y+3=0$ and P(1,2) is such that PA=PB. Then the mid point of AB is
 1) $\left(\frac{-1}{5}, \frac{13}{5} \right)$ 2) $\left(\frac{-7}{5}, \frac{9}{5} \right)$ 3) $\left(\frac{7}{5}, \frac{-9}{5} \right)$ 4) $\left(\frac{-7}{5}, \frac{-9}{5} \right)$

34. A line passing through the points (7,2), (-3,2) then the image of the line in x-axis is
 1) $y=4$ 2) $y=9$ 3) $y=-1$ 4) $y=-2$
35. Image of the curve $x^2+y^2=1$ in the line $x+y=1$ is
 1) $x^2+y^2+2x+2y+1=0$
 2) $x^2+y^2-2x+2y+1=0$
 3) $x^2+y^2+2x-2y+1=0$
 4) $x^2+y^2-2x-2y+1=0$

36. Image of (1,2) w.r.t. (-2,-1) is
1) (0,5) 2) (-4,-3) 3) (-5,-4) 4) (-4,-5)

37. The image of the point (-2,-7) under the transformation $(x,y) \rightarrow (x-2y, -3x+y)$ is

- 1) (-12,1) 2) (12,-1) 3) (-12,-1) 4) (12,1)

CENTROID, CIRCUMCENTRE, ORTHOCENTRE AND INCENTRE

38. The algebraic sum of the perpendicular distances from the vertices of a triangle to a variable line is 'O', then the line passes through the ----- of the triangle

- 1) Incentre 2) Centroid
3) Orthocentre 4) Circumcentre

39. A(1,-1) B(4,-1) C(4,3) are the vertices of a triangle. Then the equation of the altitude through the vertex 'A' is

- 1) $x = 4$ 2) $y = 4$ 3) $y + 1 = 0$ 4) $x = 1$

40. The equations of the sides of a triangle are $x-3y=0$, $4x+3y=5$, $3x+y=0$. The line $3x-4y=0$ passes through

- 1) Incentre 2) Centroid
3) Orthocentre 4) Circumcentre

41. Equation of a diameter of the circum circle of the triangle formed by the lines $3x+4y-7=0$, $3x-y+5=0$ and $8x-6y+1=0$ is

- 1) $3x-y-5=0$ 2) $3x+y+5=0$
3) $3x-y+5=0$ 4) $3x+y-5=0$

42. The incentre of the triangle formed by the lines $x \cos \alpha + y \sin \alpha = \pi$, $x \cos \beta + y \sin \beta = \pi$, $x \cos \gamma + y \sin \gamma = \pi$ is (α, β) then $\alpha + \beta =$

- 1) 0 2) 1 3) 2 4) 4

LEVEL-I (C.W)-KEY

- 1) 4 2) 3 3) 1 4) 2 5) 1 6) 2
7) 2 8) 4 9) 3 10) 2 11) 3 12) 3
13) 4 14) 3 15) 3 16) 2 17) 2 18) 2
19) 2 20) 3 21) 1 22) 1 23) 2 24) 2
25) 4 26) 1 27) 2 28) 2 29) 2 30) 1
31) 4 32) 3 33) 1 34) 4 35) 4 36) 3
37) 2 38) 2 39) 3 40) 3 41) 3 42) 1

LEVEL-I (C.W)-HINTS

- Slopes of the parallel lines are equal
- $a = -11/2$, $b = 5/3$
- let the given sides are AB, BC, CD
 $AB \parallel CD \Rightarrow b = 6$; $AB \perp BC \Rightarrow a = 3$
- $y = m(x-8)$, $m = \tan 150$
- Write the equation to a line perpendicular to $7x - y + 8 = 0$ and sub $(-4, 5)$

- Verification
- Two lines parallel to any given line make intercept of same length k between the axes in opposite quadrants

8. $m = -1$

9. $p = 2$, $x \cos \alpha + y \sin \alpha = 2$

10. $(x_1, y_1) = (3, 2)$; $\tan \theta = \frac{3}{4}$, $\sin \theta = \frac{3}{5}$, $\cos \theta = \frac{4}{5}$;

$(x, y) = (x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$

11. Eq. of the given line is $2x+y=4$ required

distance = $\frac{4}{\sqrt{5}}$

12. Distance between two parallel tangents of a circle = Diameter

13. Find the distance between $x + 7y - 3\sqrt{2} = 0$

and $x + 7y + 7\sqrt{2} = 0$

14. Verification

15. $\left| \frac{ax_1 + by_1 + c}{a \cos \alpha + b \sin \alpha} \right|$

16. Use $m : n = -L_{11} : L_{22}$

17. line parallel to $\overline{BC}$ is $x+y=k$ from given data

$\frac{|k|}{\sqrt{2}} = \frac{1}{2} \Rightarrow k = -\frac{1}{\sqrt{2}}$

($\because$ O, B are opposite sides of the line)

18. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$ $a+b+c=0$

19. $\begin{vmatrix} 3 & 2 & -5 \\ 2 & -5 & 3 \\ 5 & b & c \end{vmatrix} = 0$

20. $L_1 + L_2 + L_3 = 0$

21. $4a+2b+c=0$ & $ax+2by+3c=0$

22. Solve $3x - 2y - 1 = 0$, $x - 3y + 9 = 0$

23. intersecting point $(14, -9)$, $m = \frac{-2}{5}$

24. $\frac{2}{\lambda} = \frac{3}{k} = \frac{4}{2}$

25. The quadrilateral formed by the lines is a rhombus

26. $\sqrt{a_1^2+b_1^2} \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} : \sqrt{a_2^2+b_2^2} \begin{vmatrix} a_1 & b_1 \\ a_3 & b_3 \end{vmatrix} : \sqrt{a_3^2+b_3^2} \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$
27. $\Delta = \frac{c^2}{2|ab|}$
28. The line perpendicular to given line is
 $4x + 3y + k = 0 \therefore \frac{k^2}{24} = 6$
29. Area = $\frac{p^2}{\sqrt{3}}$ where p is height of Δ
30. Adjacent sides are perpendicular and distance between parallel sides are not equal.
31. Given equation of the diagonals of a parallelogram are at right angle. Hence it is a rhombus.
32. Mid point because $OA = OB$
33. Apply foot of the perpendicular formula
34. Line equation $y=2$ Image with respect to x-axis is $y=-2$
35. Image of $(0,0)$ in line is $(1,1)$
 $\therefore$ image circle is $(x-1)^2 + (y-1)^2 = 1$
36. The image of (x, y_1) w.r.to (x, y) is $(2x-x_1, 2y-y_1)$
37. apply given condition
38. Algebraic sum of the distances from the three non collinear points to variable line is zero then the line passing through centroid of the triangle formed by this points.
39. $AB \perp BC$
40. Given lines form a right angle triangle
41. Hypotenous is diameter
42. $(0,0)$ is equidistance from sides



LEVEL - I (H.W)

SLOPE OF A LINE

1. If the slope of a line is $\frac{-1}{\sqrt{3}}$ then its inclination is
 1) $\frac{\pi}{3}$ 2) $\frac{5\pi}{6}$ 3) $\frac{2\pi}{3}$ 4) $\frac{3\pi}{4}$
2. If the straight line $(3x+4y+5)+k(x+2y-3)=0$ is parallel to x-axis then the value of k is
 1) 1 2) -3 3) 4 4) 2
3. Number of straight lines passing through $(1,3), (7,-3), (5,-1), (6,-2)$ is
 1) 2 2) $4c_2$ 3) $4p_2$ 4) $4c_4$

SLOPE-INTERCEPT FORM, SLOPE-POINT FORM AND TWO-POINT FORM

4. The equation of the horizontal line passing through the point $(4,-7)$ is
 1) $y-7=0$ 2) $y+7=0$ 3) $y-4=0$ 4) $y+4=0$
5. The equation of the straight line making an intercept of 3 units on the y-axis and inclined at 45° to the x-axis is
 1) $y = x-1$ 2) $y = x+3$
 3) $y = 45x + 3$ 4) $y = x+45$

INTERCEPTS AND INTERCEPT FORM

6. Equation of the line having intercepts a, b on the axes such that $a+b=5$ and $ab = \frac{21}{4}$ is
 1) $3x+2y=6$ 2) $2x+3y=6$
 3) $14x+6y=21$ 4) $x+4y=4$
7. x intercept of the line parallel to $4x+7y=9$ and passing through $(2,3)$ is
 1) $\frac{25}{4}$ 2) $\frac{17}{4}$ 3) $\frac{29}{4}$ 4) $\frac{29}{7}$
8. A straight line meet the axes in A and B such that the centroid of triangle OAB is (a,a) . Then the equation of the line AB is
 1) $x+y=a$ 2) $x-y=3a$ 3) $x+y=2a$ 4) $x+y=3a$

NORMAL FORM AND SYMMETRIC FORM

9. Equation of the line on which the length of the perpendicular from origin is 5 and the angle which this perpendicular makes with the x axis is 60°
 1) $x + \sqrt{3}y = 12$ 2) $\sqrt{3}x + y = 10$
 3) $x + \sqrt{3}y = 8$ 4) $x + \sqrt{3}y = 10$
10. A point on line $x - y + 1 = 0$ at a distance $2\sqrt{2}$ from the point $(1,2)$ is
 1) $(3,4)$ 2) $(3,0)$ 3) $(-1,4)$ 4) $(0,1)$

PROBLEMS ON DISTANCES

11. The perpendicular distance from $(1,2)$ to the straight line $12x+5y=7$ is
 1) $15/13$ 2) $12/13$ 3) $5/13$ 4) $7/13$
12. The vertices of a triangle are $A(5,6)$ $B(1,-4)$ $C(-4,0)$ then the length of the altitude through the vertex A is
 1) $\frac{66}{\sqrt{41}}$ 2) $\frac{55}{\sqrt{41}}$ 3) $\frac{17}{5}$ 4) $\frac{19}{5}$

13. The distance between the parallel lines $8x+6y+5=0$ and $4x+3y-25=0$ is

1) $\frac{7}{2}$ 2) $\frac{9}{2}$ 3) $\frac{11}{2}$ 4) $\frac{5}{4}$

14. The equation of the line which is parallel to $5x+12y+1=0$ and $5x+12y+7=0$ and lying midway between them is

1) $5x+12y+13=0$ 2) $5x+12y-4=0$
3) $5x+12y+4=0$ 4) $5x+12y-6=0$

15. The point on the line $x+y=4$ that lies at a unit distance from the line $4x+3y-10=0$ is

1) (1,3) 2) (-7,11) 3) (11,-7) 4) (2,2)

POSITION OF A POINT (S) W.R.T. LINE (S)

16. The ratio in which the line $3x+4y-7=0$ divides the line joining the points (1,2) (2,3) is

1) 4:11 Internally 2) 4:11 Externally
3) 7:11 Internally 4) 7:11 Externally

17. The line segment joining the points (1,2) and (k,1) is divided by the line

$3x+4y-7=0$ in the ratio 4:9 then k is 1) 2
2) -2 3) 3 4) -3

POINT OF INTERSECTION OF LINES AND CONCURRENCY OF LINES : PP

18. If the point of intersection of $kx+4y+2=0$, $x-3y+5=0$ lies on $2x+7y-3=0$ then k=

1) 2 2) 3 3) -2 4) -3

19. The lines $px+qy+r=0$, $qx+ry+p=0$ $rx+py+q=0$, are concurrent then

1) $p+q+r=0$ 2) $p^3+q^3+r^3=3pqr$
3) $p^2+q^2+r^2-pq-qr-rp=0$
4) All the above

20. The point of concurrence of the lines

$\frac{x}{3}+\frac{y}{4}=1$, $\frac{x}{4}+\frac{y}{3}=1$, $x=y$ is

1) $\left(\frac{4}{3}, \frac{4}{3}\right)$ 2) $\left(\frac{2}{7}, \frac{2}{7}\right)$ 3) $\left(\frac{12}{7}, \frac{12}{7}\right)$ 4) $\left(\frac{7}{12}, \frac{7}{12}\right)$

21. If $4a+5b+6c=0$ then the set of lines $ax+by+c=0$ are concurrent at the point

1) $\left(\frac{2}{3}, \frac{5}{6}\right)$ 2) $\left(\frac{1}{3}, \frac{1}{3}\right)$ 3) $\left(\frac{1}{2}, \frac{4}{3}\right)$ 4) $\left(\frac{1}{3}, \frac{7}{3}\right)$

22. Let a and b be nonzero reals. Then the equation of the line passing through the origin and the point of intersection of

$x/a+y/b=1$ and $x/b+y/a=1$

1) $ax+by=0$ 2) $bx+ay=0$
3) $y-x=0$ 4) $x+y=0$

23. The line which is concurrent with the lines $2x+3y=7$, $2x=3y+1$ and passing through the origin is

1) $x+2y=0$ 2) $x-2y=0$
3) $2x+y=0$ 4) $2x-y=0$

ANGLE BETWEEN LINES

24. If θ is an acute angle between the lines $y=2x+3$, $y=x+1$ then the value of $\tan \theta =$

1) $\frac{2}{3}$ 2) $\frac{1}{3}$ 3) $\frac{3}{4}$ 4) $\frac{1}{2}$

25. The angle between the lines $kx+y+9=0$, $y-3x=4$ is 45° then the value of k is (EAM- 2007)

1) 2 or $\frac{1}{2}$ 2) 2 or $-1/2$
3) -2 or $\frac{1}{2}$ 4) -2 or $-1/2$

TRIANGLES AND AREA OF THE TRIANGLE

26. The straight lines $x+y=0$, $3x+y-4=0$ and $x+3y-4=0$ form a triangle which is

1) isosceles 2) right angled
3) equilateral 4) right angled isosceles triangle

27. If a, c, b are three terms of a G.P., then the line $ax+by+c=0$

1) has a fixed direction
2) always passes through a fixed point
3) forms a triangle with the axes whose area is constant
4) always cuts intercepts on the axes such that their sum is zero

28. Area enclosed by the co-ordinate axes and the line passing through the points (8,-3), (-4,12) is

1) $\frac{98}{5}$ 2) $\frac{49}{5}$ 3) $\frac{24}{25}$ 4) $\frac{17}{8}$

29. A straight line L is perpendicular to the line $4x-2y=1$ and forms a triangle of area 4 sq. units with coordinate axes then, an equation of L is

1) $2x+4y+8=0$ 2) $2x-4y+8=0$
3) $2x+4y+7=0$ 4) $4x-2y-7=0$

QUADRILATERALS AND AREA OF THE QUADRILATERALS

30. A square of area 25 sq.units is formed by taking two sides as $3x + 4y = k_1$ and $3x + 4y = k_2$ then $|k_1 - k_2| =$
 1) 5 2) 1 3) 25 4) 125
31. The equations of two sides of a square whose area is 25 sq.units are $3x - 4y = 0$ and $4x + 3y = 0$. The equation of other two sides of the square are
 1) $3x - 4y \pm 25 = 0, 4x + 3y \pm 25 = 0$
 2) $3x - 4y \pm 5 = 0, 4x + 3y \pm 5 = 0$
 3) $3x - 4y \pm 5 = 0, 4x + 3y \pm 25 = 0$
 4) $3x - 4y = 0, 4x + 3y = 0$

Foot and Image:

32. If $2x + 3y = 5$ is the perpendicular bisector of the line segment joining the points $A(1, \frac{1}{3})$ and B then B = (EAM- 2013)
 1) $(\frac{21}{13}, \frac{49}{39})$ 2) $(\frac{17}{13}, \frac{31}{39})$
 3) $(\frac{7}{13}, \frac{49}{39})$ 4) $(\frac{21}{13}, \frac{31}{39})$
33. If $(2, -3)$ is the foot of the perpendicular from $(-4, 5)$ on a line then its equation is
 1) $3x - 4y + 28 = 0$ 2) $3x - 4y - 18 = 0$
 3) $3x - 4y + 18 = 0$ 4) $3x - 4y - 17 = 0$
34. If $(-2, 6)$ is the image of the point $(4, 2)$ with respect to the line $L = 0$, then $L =$ (EAM- 2014)
 1) $6x - 4y - 7 = 0$ 2) $2x + 3y - 5 = 0$
 3) $3x - 2y + 5 = 0$ 4) $3x - 2y + 10 = 0$

CENTROID, CIRCUMCENTRE, ORTHOCENTRE AND INCENTRE

37. The vertices of a triangle are $(2, 0)$ $(0, 2)$ $(4, 6)$ then the equation of the median through the vertex $(2, 0)$ is
 1) $x + y - 2 = 0$ 2) $x = 2$
 3) $x + 2y - 2 = 0$ 4) $2x + y - 4 = 0$

38. If the algebraic sum of the perpendicular distances from the points $(2, 0)$ $(0, 2)$ and $(4, 4)$ to a variable line is '0', then the line passes through the fixed point
 1) $(1, 1)$ 2) $(3, 3)$ 3) $(2, 2)$ 4) $(0, 0)$
39. If the equations of the three sides of a triangle are $x + y = 1$, $3x + 5y = 2$ and $x - y = 0$ then the orthocentre of the triangle lies on the line
 1) $5x - 3y = 2$ 2) $3x - 5y - 1 = 0$
 3) $2x - 3y = 1$ 4) $5x - 3y = 1$
40. Circum centre of the triangle formed by the lines $x + y = 0$, $2x + y + 5 = 0$, $x - y = 2$ is
 1) $(-2, -1)$ 2) $(-3, 1)$ 3) $(-4, 3)$ 4) $(-1, -3)$
41. The incentre of the triangle formed by the lines $3x + 4y = 10$, $5x + 12y = 26$, $7x + 24y = 50$ is (α, β) then $\alpha + \beta =$
 1) 0 2) 1 3) 2 4) 4

LEVEL-I(H.W) - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 2 | 2) 2 | 3) 4 | 4) 2 | 5) 2 | 6) 3 |
| 7) 3 | 8) 4 | 9) 4 | 10) 1 | 11) 1 | 12) 1 |
| 13) 3 | 14) 3 | 15) 2 | 16) 2 | 17) 2 | 18) 2 |
| 19) 4 | 20) 3 | 21) 1 | 22) 3 | 23) 2 | 24) 2 |
| 25) 2 | 26) 3 | 27) 3 | 28) 1 | 29) 1 | 30) 3 |
| 31) 1 | 32) 1 | 33) 2 | 34) 3 | 35) 2 | 36) 4 |
| 37) 2 | 38) 3 | 39) 4 | 40) 2 | 41) 1 | |

LEVEL-I(H.W) - HINTS

- $\tan \theta = \frac{-1}{\sqrt{3}}$
- Coefficient of $x = 0$
- Given points are collinear
- $y = y_1$
- $y = mx + c$
- $a = 3/2, b = 7/2$
- $y - 3 = \frac{-4}{7}(x - 2)$, Put $y = 0$
- Intercepts α, β ; $\therefore \left(\frac{\alpha}{3}, \frac{\beta}{3}\right) = (a, a)$
- $P = 5, \alpha = 60^\circ$; $x \cos \alpha + y \sin \alpha = P$
- $P(x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$ with $r = 2\sqrt{2}, \theta = \pi/4$
- The perpendicular distance from (x_1, y_1) to the st line $ax + by + c = 0$ is $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

12. Find the perpendicular distance from A to BC

$$13. d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

$$14. ax + by + \frac{c_1 + c_2}{2} = 0$$

15. Verification

$$16. \text{Use } m : n = -L_{11} : L_{22}$$

$$17. -\frac{L_{11}}{L_{22}} = \frac{4}{9} \Rightarrow 3 - 3k = 9$$

$$18. \begin{vmatrix} k & 4 & 2 \\ 1 & -3 & 5 \\ 2 & 7 & -3 \end{vmatrix} = 0$$

$$19. \begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} = 0$$

$$(p + q + r)(p^2 + q^2 + r^2 - pq - qr - rp) = 0$$

$$20. \text{Point of intersection is } \left(\frac{12}{7}, \frac{12}{7}\right)$$

$$21. a\left(\frac{4}{6}\right) + b\left(\frac{5}{6}\right) + c = 0$$

$$22. \text{Intersecting point of } \frac{x}{a} + \frac{y}{b} = 1 \text{ and}$$

$$\frac{x}{b} + \frac{y}{a} = 1 \text{ is } \left(\frac{ab}{a+b}, \frac{ab}{a+b}\right)$$

23. point of intersection of given lines is (2,1)

$$24. \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$25. \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = 1$$

$$26. \sqrt{1+1} \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} : \sqrt{3^2+1^2} \begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix} : \sqrt{1^2+3^2} \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix}$$

$$27. c^2 = ab; \Delta = \frac{c^2}{2ab} = \frac{1}{2}$$

$$28. \text{The line equation is } 5x + 4y - 28 = 0, \Delta = \frac{c^2}{2|ab|}$$

29. $2x + y + k = 0$ forms a triangle of area 4.

$$30. \left| \frac{k_1 - k_2}{5} \right| = 5 \Rightarrow |k_1 - k_2| = 25$$

31. Verification

32. B is image of A

33. Line is perpendicular to AB

34. Required line is the perpendicular bisector of given points.

35. Image of A w.r.to diagonal BD is C

36. The reflection of the point (a,b) in the line $x=y$ is (b,a)

37. $A(2,0)B(0,2)C(4,6)$; mid point of BC is $D(2,4)$; Equation of AD is $x = 2$

38. Centroid

39. given lines form a right angled triangle

40. Given lines form a right angle triangle

41. Incentre is equidistance from sides,
Hence $I = (0, 0)$



LEVEL - II (C.W)

SLOPE OF A LINE

1. The lines $p(p^2 + 1)x - y + q = 0$ and

$$(p^2 + 1)^2 x + (p^2 + 1)y + 2q = 0 \text{ are}$$

perpendicular to a common line for

[AIEEE - 2009]

- 1) exactly one value of p
- 2) exactly two values of p
- 3) more than two values of p
- 4) no values of p

2. The slope of the line passing through the points $(2, \sin \theta), (1, \cos \theta)$ is 0 then general solution of θ

- 1) $n\pi + \frac{\pi}{4}, \forall n \in \mathbb{Z}$
- 2) $n\pi - \frac{\pi}{4}, \forall n \in \mathbb{Z}$
- 3) $n\pi \pm \frac{\pi}{4}, \forall n \in \mathbb{Z}$
- 4) $n\pi, \forall n \in \mathbb{Z}$

SLOPE-INTERCEPT FORM, SLOPE-POINT FORM AND TWO-POINT FORM

3. The perpendicular bisector of the line segment joining $P(1, 4)$ and $Q(K, 3)$ has Y intercept -4. then a possible value of K is

(AIEEE-2008)

- 1) -4
- 2) 1
- 3) 2
- 4) -2

4. $P(\alpha, \beta)$ lies on the line $y=6x-1$ and $Q(\beta, \alpha)$ lies on the line $2x-5y=5$. Then the equation of the line $\overline{PQ}$ is
 1) $2x+y=3$ 2) $3x+2y=5$ 3) $x+y=6$ 4) $3x+y=7$
5. A line joining $A(2,0)$ and $B(3,1)$ is rotated about A in anticlock wise direction through angle 15° , then the equation of AB in the new position is
 1) $y = \sqrt{3} x - 2$ 2) $y = \sqrt{3} (x - 2)$
 3) $y = \sqrt{3} (x + 2)$ 4) $x - 2 = \sqrt{3} y$

INTERCEPTS AND INTERCEPT FORM

6. The line $2x+3y=6$, $2x+3y=8$ cut the X -axis at A, B respectively. A line $L=0$ drawn through the point $(2,2)$ meets the X -axis at C in such a way that abscissa of A, B, C are in arithmetic Progression. then the equation of the line L is
 1) $2x+3y=10$ 2) $3x+2y=10$
 3) $2x-3y=10$ 4) $3x-2y=10$
7. The sum of the intercepts cut off by the axes on lines
 $x + y = a, x + y = ar, x + y = ar^2, \dots$
 where $a \neq 0$ and $r = \frac{1}{2}$
 1) $2a$ 2) $a\sqrt{2}$ 3) $2\sqrt{2}a$ 4) a
8. The equation of the straight line which bisects the intercepts between the axes of the lines $x + y = 2$ and $2x + 3y = 6$ is
 1) $2x = 3$ 2) $y = 1$ 3) $2y = 3$ 4) $x = 1$
9. Equation of the line passing through $(0, 1)$ and having intercepts in the ratio $2 : 3$ is
 1) $2x + 3y = 3$ 2) $2x - 3y + 3 = 0$
 3) $3x + 2y = 2$ 4) $2x - 3y - 3 = 0$

NORMAL FORM AND SYMMETRIC FORM

10. A straight line is such that its distance of 5 units from the origin and its inclination is 135° . The intercepts of the line on the co-ordinate axes are
 1) $5, 5$ 2) $\sqrt{2}, \sqrt{2}$
 3) $5\sqrt{2}, 5\sqrt{2}$ 4) $5/\sqrt{2}, 5/\sqrt{2}$
11. Angles made with the x - axis by two lines drawn through the point $(1, 2)$ and cutting the line $x + y = 4$ at a distance $\sqrt{\frac{2}{3}}$ from the point $(1, 2)$ are

- 1) $\frac{\pi}{6}$ and $\frac{\pi}{3}$ 2) $\frac{\pi}{8}$ and $\frac{3\pi}{8}$
 3) $\frac{\pi}{12}$ and $\frac{5\pi}{12}$ 4) $\frac{\pi}{4}$ and $\frac{\pi}{2}$

PROBLEMS ON DISTANCES

12. Perpendicular distance from the origin to the line joining the points $(a \cos \theta, a \sin \theta)$ $(a \cos \phi, a \sin \phi)$ is
 1) $2a \cos (\theta - \phi)$ 2) $a \cos \left(\frac{\theta - \phi}{2} \right)$
 3) $4a \cos \left(\frac{\theta - \phi}{2} \right)$ 4) $a \cos \left(\frac{\theta + \phi}{2} \right)$
13. One side of an equilateral triangle is $3x+4y=7$ and its vertex is $(1, 2)$. Then the length of the side of the triangle is
 1) $\frac{4\sqrt{3}}{17}$ 2) $\frac{3\sqrt{3}}{16}$ 3) $\frac{8\sqrt{3}}{15}$ 4) $\frac{4\sqrt{3}}{15}$
14. Equation of the line through the point of intersection of the lines $3x+2y+4=0$ and $2x+5y-1=0$ whose distance from $(2, -1)$ is 2.
 1) $2x-y+5=0$ 2) $4x+3y+5=0$
 3) $x+2=0$ 4) $3x+y+5=0$
15. If p, q denote the lengths of the perpendiculars from the origin on the lines $x \sec \alpha - y \csc \alpha = a$ and $x \cos \alpha + y \sin \alpha = a \cos 2\alpha$ then (Eam 2013)
 1) $4p^2 + q^2 = a^2$ 2) $p^2 + q^2 = a^2$
 3) $p^2 + 2q^2 = a^2$ 4) $4p^2 + q^2 = 2a^2$
16. The distance between two parallel lines is $p^1 - p$ equation of one line is $x \cos \alpha + y \sin \alpha = p$ then the equation of the 2^{nd} line is
 1) $x \cos \alpha + y \sin \alpha + p^1 + 2p = 0$
 2) $x \cos \alpha + y \sin \alpha = 2p^1 - p$
 3) $x \cos \alpha + y \sin \alpha = 0$
 4) $x \cos \alpha + y \sin \alpha + p^1 - 2p = 0$
17. The ratio in which the line $3x+4y+2=0$ divides the distance between $3x+4y+5=0$ and $3x+4y-5=0$ is
 1) $7 : 3$ 2) $3 : 7$ 3) $2 : 3$ 4) $3 : 4$
18. The equations of the lines parallel to $4x + 3y + 2 = 0$ and at a distance of '4' units from it are
 1) $4x + 3y + 22 = 0, 4x + 3y - 20 = 0$
 2) $4x + 3y + 22 = 0, 4x + 3y - 18 = 0$
 3) $4x + 3y - 18 = 0, 4x + 3y - 20 = 0$
 4) $4x - 3y - 18 = 0, 4x + 3y - 20 = 0$

POSITION OF A POINT (S) W.R.T. LINE (S)

19. The range of α for which the points $(\alpha, \alpha+2)$ and $\left(\frac{3\alpha}{2}, \alpha^2\right)$ lie on opposite sides of the line $2x+3y-6=0$

- 1) $(-\infty, -2)$ 2) $(0, 1)$
3) $(-\infty, -2) \cup (0, 1)$ 4) $(-\infty, 1) \cup (2, \infty)$

20. If $P\left(1 + \frac{t}{\sqrt{2}}, 2 + \frac{t}{\sqrt{2}}\right)$ be any point on a line then the range of values of t for which the point P lies between the parallel lines $x+2y=1$ and $2x+4y=15$ is

- 1) $-\frac{4\sqrt{2}}{5} < t < \frac{5\sqrt{2}}{6}$ 2) $-\frac{4\sqrt{2}}{3} < t < \frac{5\sqrt{2}}{6}$
3) $t < \frac{-4\sqrt{2}}{3}$ 4) $t < \frac{5\sqrt{2}}{6}$

21. A point which lies between $2x+3y-7=0$ and $2x+3y+12=0$ is

- 1) $(5, 1)$ 2) $(-1, 3)$ 3) $(3, -5)$ 4) $(7, -1)$

22. A line L cuts the sides AB, BC of $\triangle ABC$ in the ratio $2 : 5, 7 : 4$ respectively. Then the line L cuts CA in the ratio

- 1) $7 : 10$ 2) $7 : -10$ 3) $10 : 7$ 4) $10 : -7$

POINT OF INTERSECTION OF LINES AND CONCURRENCY OF LINES

23. The number of integral values of m for which x -coordinate of point of intersection of the lines $3x+4y=9$ and $y=mx+1$ is also an integer is

- 1) 2 2) 0 3) 4 4) 11

24. The line parallel to the x -axis and passing through the intersection of the lines $ax+2by+3b=0$ and $bx-2ay-3a=0$, where $(a, b) \neq (0, 0)$ is

- (1) Above the x -axis at a distance of $3/2$ from it
(2) Above the x -axis at a distance of $2/3$ from it
(3) Below the x -axis at a distance of $3/2$ from it
(4) Below the x -axis at a distance of $2/3$ from it

25. If a, b, c form a G.P. with common ratio r , the sum of the ordinates of the points of intersection of the line $ax+by+c=0$ and the curve $x+2y^2=0$ is

- 1) $\frac{-r}{2}$ 2) $\frac{-r^2}{2}$ 3) $\frac{r}{2}$ 4) $\frac{r^2}{2}$

26. Consider a family of straight lines

$(x+y) + \lambda(2x-y+1) = 0$. Find the equation of the straight line belonging to this family that is farthest from $(1, -3)$.

- 1) $3x-3y+2=0$ 2) $6x+15y-7=0$
3) $5x+2y+1=0$ 4) $6x-15y+7=0$

27. If the line $x=a+m$, $y=-2$ and $y=mx$ are concurrent, then least value of $|a|$ is

- 1) 0 2) $\sqrt{2}$ 3) $2\sqrt{2}$ 4) 2

28. If $a \neq b \neq c$, if $ax+by+c=0$

$bx+cy+a=0$ and $cx+ay+b=0$ are concurrent. Then the value of

$$2^{a^2b^{-1}c^{-1}} 2^{b^2c^{-1}a^{-1}} 2^{c^2a^{-1}b^{-1}}$$

- 1) 1 2) 4 3) 8 4) 16

29. Line $ax+by+p=0$ makes angle $\pi/4$ with

$x \cos \alpha + y \sin \alpha = p$, $p \in R^+$. If these lines

and the line $x \sin \alpha - y \cos \alpha = 0$ are concurrent, then

- 1) $a^2+b^2=1$ 2) $a^2+b^2=2$
3) $2(a^2+b^2)=1$ 4) $a^2-b^2=1$

ANGLE BETWEEN LINES

30. If p, q, r are distinct, then $(q-r)x + (r-p)y + (p-q)=0$ and $(q^3-r^3)x + (r^3-p^3)y + (p^3-q^3)=0$ represents the same line if

- 1) $p+q+r=0$ 2) $p=q=r$
3) $p^2+q^2+r^2=0$ 4) $p^3+q^3+r^3=0$

31. The lines $(a+b)x + (a-b)y - 2ab = 0$,

$(a-b)x + (a+b)y - 2ab = 0$ and $x+y=0$ form an isosceles triangle whose vertical angle is

- 1) $\frac{\pi}{2}$ 2) $\tan^{-1}\left(\frac{2ab}{a^2-b^2}\right)$
3) $\tan^{-1}\left(\frac{a}{b}\right)$ 4) $2 \tan^{-1}\left(\frac{a}{b}\right)$

32. If $2(\sin a + \sin b)x - 2\sin(a-b)y = 3$ and $2(\cos a + \cos b)x + 2\cos(a-b)y = 5$ are perpendicular then $\sin 2a + \sin 2b =$
- $\sin(a-b) - 2\sin(a+b)$
 - $\sin 2(a-b) - 2\sin(a+b)$
 - $2\sin(a-b) - \sin(a+b)$
 - $\sin 2(a-b) - \sin(a+b)$

33. Two equal sides of an isosceles triangle are given by $7x - y + 3 = 0$ and $x + y - 3 = 0$ and the third side passes through the point $(1, 10)$ then the slope m of the third side is given by

- $3m^2 - 1 = 0$
- $m^2 + 1 = 0$
- $3m^2 + 8m - 3 = 0$
- $m^2 - 3 = 0$

34. The diagonal of a square is $8x - 15y = 0$ and one vertex of the square is $(1, 2)$. Then the equations to the sides of the square passing through the vertex are

- $22x + 8y = 9, 22x - 8y = 52$
- $23x + 7y = 9, 7x - 23y = 52$
- $23x - 7y = 9, 7x + 23y = 53$
- $22x - 8y = 9, 22x + 8y = 52$

TRIANGLES AND AREA OF THE TRIANGLE

35. Area of triangle formed by angle bisectors of coordinate axes and the line $x=6$ in sq. units is
- 36
 - 18
 - 72
 - 9

36. The quadratic equation whose roots are the x and y intercepts of the line passing through $(1,1)$ and making a triangle of area A with the co-ordinate axes is

- $x^2 + Ax + 2A = 0$
- $x^2 - 2Ax + 2A = 0$
- $x^2 - Ax + 2A = 0$
- $(x - A)(x + A) = 0$

37. A line passing through $(3,4)$ meets the axes $\overline{OX}$ and $\overline{OY}$ at A and B respectively. The minimum area of the triangle OAB in square units is

- 8
- 16
- 24
- 32

QUADRILATERALS AND AREA OF THE QUADRILATERALS

38. The figure formed by the straight lines $\sqrt{3}x + y = 0$, $\sqrt{3}y + x = 0$, $\sqrt{3}x + y = 1$, $\sqrt{3}y + x = 1$ is

- a rectangle
- a square
- a rhombus
- parallelogram

39. Let the base of a triangle lie along the line $x = a$ and be of length a . The area of this triangle is a^2 , if the vertex lies on the line

- $x + a = 0$
- $x = 0$
- $2x - a = 0$
- $x - a = 0$

40. The area bounded by $y = |x| - 1$, $y = -|x| + 1$

- 1
- 2
- $2\sqrt{2}$
- 4

41. The area enclosed by $2|x| + 3|y| \leq 6$ is

- 3 sq. units
- 4 sq. units
- 12 sq. units
- 24 sq. units

FOOT AND IMAGE

42. The point on the line $3x - 2y = 1$ which is closest to the origin is

- $\left(\frac{3}{13}, \frac{2}{13}\right)$
- $\left(\frac{5}{11}, \frac{2}{11}\right)$
- $\left(\frac{3}{5}, \frac{2}{5}\right)$
- $\left(\frac{3}{13}, \frac{-2}{13}\right)$

43. The reflection of $y = \sqrt{x}$ w.r.t. y -axis is

- $y = -\sqrt{x}$
- $y = \sqrt{-x}$
- $y = -\sqrt{-x}$
- $x = \sqrt{y}$

44. The points $(-1, 1)$ and $(1, -1)$ are symmetrical about the line

- $y + x = 0$
- $y = x$
- $x + y = 1$
- $x - y = 1$

45. The equation of perpendicular bisectors of sides AB, BC of $\triangle ABC$ are $x - y - 5 = 0$, $x + 2y = 0$ respectively and $A(1, -2)$ then coordinate of C are

- $(1, 0)$
- $(0, 1)$
- $(5, 0)$
- $(0, 0)$

CENTROID, CIRCUMCENTRE, ORTHOCENTRE AND INCENTRE

46. If one vertex of an equilateral triangle is the origin and side opposite to it has the equation $x + y = 1$, then the orthocentre of the triangle is

- $\left(\frac{1}{3}, \frac{1}{3}\right)$
- $\left(\frac{2}{3}, \frac{2}{3}\right)$
- $(1, 1)$
- $(1, 3)$

47. If the circum centre of the triangle lies at $(0, 0)$ and centroid is middle point of $(a^2 + 1, a^2 + 1)$ and $(2a, -2a)$ then the orthocentre lies on

- $(a-1)^2 x - (a+1)^2 y = 0$
- $(a-1)^2 x + (a+1)^2 y = 0$
- $(a-1)^2 x + (a+1)^2 y + 56 = 0$
- $(a-1)^2 x + (a+1)^2 y - 56 = 0$

48. The orthocentre of the triangle formed by the lines $x+y=1$, $2x+3y=6$ and $4x-y+9=0$ lies in quadrant number

- 1) 1st 2) IIrd 3) IIIrd 4) IVth

49. If the straight lines $2x+3y-1=0$, $x+2y-1=0$ and $ax+by-1=0$ form a triangle with origin as orthocentre, then (a,b) is given by

- 1) $(6,4)$ 2) $(-3,3)$ 3) $(-8,8)$ 4) $(0,7)$

50. In $\triangle ABC$, equation to AB is $2x+3y-5=0$, altitude through A is $x-y+4=0$ and altitude through B is $2x-y-1=0$. Then the vertex C is

- 1) $\left(-\frac{1}{5}, \frac{9}{5}\right)$ 2) $\left(\frac{1}{5}, \frac{9}{5}\right)$ 3) $\left(\frac{1}{5}, -\frac{9}{5}\right)$ 4) $\left(-\frac{1}{5}, -\frac{9}{5}\right)$

51. Centroid of the triangle, formed by the lines $x+2y-5=0$, $2x+y-7=0$, $x-y+1=0$ is

- 1) $(1,3)$ 2) $(3,5)$ 3) $(2,2)$ 4) $(1,1)$

ANGULAR BISECTORS

52. The acute angle bisector between the lines $3x-4y-5=0$, $5x+12y-26=0$ is

- 1) $7x-56y+32=0$ 2) $9x-3y+13=0$
3) $14x-112y+65=0$ 4) $7x-13y+9=0$

53. The equation of the bisector of the angle between the lines $x-7y+5=0$, $5x+5y-3=0$ which is the supplement of the angle containing the origin will be

- 1) $x+3y-2=0$ 2) $x-3y+2=0$
3) $3x-y+1=0$ 4) $3x+y+2=0$

54. Reflection of $3x+4y+5=0$ w.r.to the line $2x+y+1=0$ is

1. $2x+1=0$ 2. $2x-1=0$
3. $5x-1=0$ 4. $5x+1=0$

55. Two sides of a Rhombus $ABCD$ are parallel to the lines $x-y=5$ and $7x-y=3$. The diagonals intersect at $(2,1)$ then the equations of the diagonals are

- 1) $x-y=1$, $7x-y=13$ 2) $x+y=3$, $x+7y=9$
3) $x+2y=4$, $2x-y=3$ 4) $3x+4y=10$, $4x-3y=5$

56. Let $P = (-1,0)$, $Q = (0,0)$ and $R = (3, 3\sqrt{3})$ be three points. Then the equation of the bisector of angle PQR is (AIEEE 2007)

- 1) $\frac{\sqrt{3}}{2}x + y = 0$ 2) $x + \sqrt{3}y = 0$

- 3) $\sqrt{3}x + y = 0$ 4) $x + \frac{\sqrt{3}}{2}y = 0$

OPTIMIZATION AND REFLECTION IN SURFACE

57. A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x -axis, the equation of the reflected ray is [JEE-MAINS 2013]

- 1) $y = x + \sqrt{3}$ 2) $\sqrt{3}y = x - \sqrt{3}$
3) $y = 3x - \sqrt{3}$ 4) $\sqrt{3}y = x - 1$

58. Consider the points $A(0,1)$ and $B(2,0)$ and P be a point on the line $4x+3y+9=0$. Coordinates of P such that $|PA-PB|$ is maximum are

- 1) $\left(\frac{-24}{5}, \frac{17}{5}\right)$ 2) $\left(\frac{-84}{5}, \frac{13}{5}\right)$
3) $\left(\frac{-6}{5}, \frac{17}{5}\right)$ 4) $(0, -3)$

MISCELLANEOUS PROBLEMS

59. A straight line which make equal intercepts on +ve x and y axes and which is at a distance '1' unit from the origin intersects the straight line $y=2x+3+\sqrt{2}$ at (x_0, y_0) then $2x_0 + y_0 =$ [EAM 2010]

- 1) $3+\sqrt{2}$ 2) $\sqrt{2}-1$ 3) 1 4) 0

60. p is the length of the perpendicular drawn from the origin upon a straight line then the locus of mid point of the portion of the line intercepted between the coordinate axes is

- 1) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{p^2}$ 2) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{2}{p^2}$

- 3) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$ 4) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{p}$

61. Equation of the line passing through the point $(2,3)$ and making intercept 2 units between the lines $y+2x=3$, $y+2x=5$ is

- 1) $x=2$ 2) $y=3$ 3) $x+y=5$ 4) $x+y=7$

62. The number of lines that can be drawn through the point $(4,-5)$ at a distance of 10 units from the point $(1,3)$ is

- 1) 0 2) 1 3) 2 4) Infinite

63. The number of circles that touch all the 3 lines $2x+y=3$, $4x-y=3$, $x+y=2$ is

- 1) 0 2) 1 3) 2 4) 4

LEVEL-II (C.W)- KEY

- 1) 1 2) 1 3) 1 4) 3 5) 2 6) 1
 7) 3 8) 2 9) 3 10) 3 11) 3 12) 2
 13) 3 14) 2 15) 1 16) 4 17) 2 18) 2
 19) 3 20) 2 21) 3 22) 4 23) 1 24) 3
 25) 3 26) 4 27) 3 28) 3 29) 2 30) 1
 31) 2 32) 2 33) 3 34) 3 35) 1 36) 2
 37) 3 38) 3 39) 1 40) 2 41) 3 42) 4
 43) 2 44) 2 45) 3 46) 1 47) 1 48) 2
 49) 3 50) 2 51) 3 52) 3 53) 1 54) 3
 55) 3 56) 3 57) 2 58) 1 59) 2 60) 3
 61) 1 62) 1 63) 2

LEVEL-II (C.W)- HINTS

- $\frac{a_1}{a_2} = \frac{b_1}{b_2}$
- $\sin \theta = \cos \theta \quad \tan \theta = 1$
- $(0, -4)$ lies on perpendicular bisector $\overline{PQ}$
- By solving $\beta = 6\alpha - 1$ and $2\beta - 5\alpha = 5$ we get $P(1, 5), Q(5, 1)$
- New position of AB makes $15^\circ + 45^\circ$ inclination with x-axis
- $A = (3, 0) B(4, 0); c = \left(2 - \frac{2}{m}, 0\right)$
- Intercepts between the axes made by the given lines are $a\sqrt{2}, ar\sqrt{2}, ar^2\sqrt{2} \dots$
- Midpoints of intercepts of given lines are $(1, 1), (3/2, 1)$
- $mx + ny = mx_1 + ny_1$
- $\alpha = 135^\circ - 90^\circ, P = 5$
- $r = \sqrt{\frac{2}{3}}, (x_1, y_1) = (1, 2)$
- Find the distance between $(0, 0)$ and midpoint of $(a \cos \theta, a \sin \theta)$ and $(a \cos \phi, a \sin \phi)$
- $\frac{\sqrt{3}}{2}a = \frac{4}{5}$
- Point of intersection $(-2, 1)$ and verification
- Put $\alpha = 45$ and use perpendicular distance formula
- Verify the distance between the parallel lines
- distance between $3x+4y+2=0$ & $3x+4y+5=0$
distance between $3x+4y+2=0$ & $3x+4y-5=0$
- $ax + by + c \pm d\sqrt{a^2 + b^2} = 0$

- Points lie on opposite sides of the line
 $\Rightarrow L_{11}L_{22} < 0$
 $\Rightarrow 5\alpha(3\alpha + 3\alpha^2 - 6) < 0 \rightarrow \alpha(\alpha + 2)(\alpha - 1) < 0$
 $\Rightarrow \alpha \in (-\infty, -2) \cup (0, 1)$
- Origin, P lies opposite side to the first line and same side to the second line
- Verify $L_p L_p^1 < 0$ for options
- $\left(\frac{BD}{DC}\right)\left(\frac{CE}{EA}\right)\left(\frac{AF}{FB}\right) = -1$
- By solving, given equations we get $x = \frac{5}{3 + 4m}$
 x is an integer of $3 + 4m = \pm 1, \pm 5$,
 $\therefore$ integral values of m are $-1, -2$
- Eq. of required line parallel to x-axis
 $\Rightarrow$ slope $= 0 \Rightarrow \lambda = -a/b$
 Equation $= 2y + 3 = 0$
- By solving two equations we get $2y^2 - ry - r^2 = 0$
- Line is perpendicular to line joining $(1, -3)$ and point of concurrency
- by eliminating x, y from three equations we get $-2 = m(a + m) \Rightarrow m^2 + am + 2 = 0$
 Since $m \in R \Rightarrow \text{disc} \geq 0$
 $\therefore a^2 - 8 \geq 0 \Rightarrow |a| \geq 2\sqrt{2}$
- $a + b + c = 0; a^3 + b^3 + c^3 = 3abc$
- $ax + by + c = 0$ is angle bisector of given two lines.
- $\frac{q^3 - r^3}{q - r} = \frac{r^3 - p^3}{r - p} = \frac{p^3 - q^3}{p - q}$
- $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
- $m_1 m_2 = -1$
- $\frac{m - 7}{1 + 7m} = -\frac{m + 1}{1 - m} \Rightarrow 3m^2 + 8m - 3 = 0$
- By using $\frac{m \pm \tan \alpha}{1 \mp m \tan \alpha}$
 required slopes are $23/7, -7/23$
- Equations of the angular bisectors of the axes are $y = x$ and $y = -x$
- $a + b = ab, \frac{1}{2}ab = A$

37. $(p, q) = (3, 4)$ then minimum area $= 2pq$
38. Adj sides are not perpendicular and $d_1 = d_2$
39. Area $= a^2 \Rightarrow$ height of $\Delta = 2a$
40. From the diagram required area
 $= 4(\text{area of } \Delta) = 4\left(\frac{1}{2}\right) = 2$
41. required area is $\frac{1}{2} \times 6 \times 4 = 12$ sq. units
42. Foot of the perpendicular
43. The image of (x, y) w.r.to y-axis is $(-x, y)$
44. Required line is perpendicular bisector of $\overline{AB}$
45. B is image of A w.r.to $x-y-5=0$
 C is image of B w.r.to $x-2y=0$
46. Foot of the perpendicular $D = \left(\frac{1}{2}, \frac{1}{2}\right)$
 G(=O) divides median in the ratio 2:1
47. Line joining circumcentre and centroid
48. apply $(a_1a_2+b_1b_2)$ $L_3 = (a_1a_3+b_1b_3)$ $L_2 = (a_2a_3+b_2b_3)$ L_1
49. Equation of AO is
 $(2x+3y-1) + \lambda(x+2y-11) = 0$ passes through
 $(0,0) \Rightarrow \lambda = -1$
 Since $AO \perp BC$ we have $a=-b$
 similarly apply $BO \perp AC$
50. Orthocentre $= (5,9)$
 Altitude through C which is perpendicular to
 $\overline{AB}$ is $3x-2y+3=0$
51. Apply $L_1\lambda_1 = L_2\lambda_2 = L_3\lambda_3$
52. $a_1a_2+b_1b_2 = -32 < 0$, $c_1c_2 = 130 > 0$
 Use $\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}} = \frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$
53. Bisector not containing origin is
 $\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}} = -\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$
 $(\because c_1, c_2 \text{ have opposite signs})$
54. $(l^2+m^2)(ax+bx+c) = 2(al+bm)(lx+my+n)$
55. Diagonals are parallel to angular bisectors.
56. T divides PR in the ratio PQ:QR=1:6
57. Slope of reflected ray is $\frac{1}{\sqrt{3}}$ and it passing through
 $(\sqrt{3}, 0)$ is $\frac{y-0}{x-\sqrt{3}} = \frac{1}{\sqrt{3}}$ $\sqrt{3}y = x - \sqrt{3}$

58. P.I of given line and AB
59. Equation of the straight line having equal intercepts is $x+y=k$ and proceed.
60. Equate the distance from $(0,0)$ to the line
 $\frac{x}{x_1} + \frac{y}{y_1} = 2$
61. Verification
62. $AB < d$
63. Given lines are concurrent.



LEVEL - II (H.W)

SLOPE OF A LINE

1. If the inclination of the line
 $(2-k)x - (1-k)y + (5-2k) = 0$ is $\frac{3\pi}{4}$ then the value of k is
 1) $\frac{5}{2}$ 2) $\frac{-3}{2}$ 3) $\frac{2}{3}$ 4) $\frac{3}{2}$
2. If the line joining the points $(at_1^2, 2at_1)$ $(at_2^2, 2at_2)$ is parallel to $y=x$ then $t_1+t_2 =$
 1) $\frac{1}{2}$ 2) 4 3) $\frac{1}{4}$ 4) 2

SLOPE-INTERCEPT FORM, SLOPE-POINT FORM AND TWO-POINT FORM

3. The line $2x+3y+12=0$ cuts the axes at A & B. Then the equation of the perpendicular bisector of $\overline{AB}$ is
 1) $3x-2y+5=0$ 2) $3x-2y+7=0$
 3) $3x-2y+9=0$ 4) $3x-2y+8=0$
4. If t_1, t_2 are roots of the equation $t^2 + \lambda t + 1 = 0$ where λ is an arbitrary constant, then the line joining the points $(at_1^2, 2at_1)$ $(at_2^2, 2at_2)$ always passes through the fixed point
 1) $(-a, 0)$ 2) $(0, a)$ 3) $(a, 0)$ 4) $(0, -a)$
5. ABCD is a parallelogram. Equations of AB and AD are $4x + 5y = 0$ and $7x + 2y = 0$ and the equation of diagonal BD is $11x + 7y + 9 = 0$. The equation of AC is
 1) $x + y = 0$ 2) $x - y = 0$
 3) $x + y + 1 = 0$ 4) $x + y - 1 = 0$

INTERCEPTS AND INTERCEPT FORM

6. The equation of the straight line whose intercepts on x-axis and y-axis are respectively twice and thrice of those by the line $3x + 4y = 12$, is
 1) $9x + 8y = 72$ 2) $9x - 8y = 72$
 3) $8x + 9y = 72$ 4) $8x + 9y + 72 = 0$
7. The equation of a straight line parallel to $2x + 3y + 11 = 0$ and which is such that the sum of its intercepts on the axes is 15.
 1) $2x + 3y = 15$ 2) $3x + 2y = 10$
 3) $2x - 3y = 10$ 4) $2x + 3y = 18$
8. The straight line through $P(1, 2)$ is such that its intercept between the axes is bisected at P. Its equation is
 1) $x + 2y = 5$ 2) $x + y - 3 = 0$
 3) $x - y + 1 = 0$ 4) $2x + y - 4 = 0$
9. If $(4, -3)$ divides the line segment between the axes in the ratio 4 : 5 then its equation is
 1) $15x + 16y - 12 = 0$ 2) $3x - 4y - 24 = 0$
 3) $15x - 16y + 108 = 0$ 4) $15x - 16y - 108 = 0$

NORMAL FORM AND SYMMETRIC FORM

10. If a line AB makes an angle θ with OX and is at a distance of p units from the origin then the equation of AB is
 1) $x \sin \theta - y \cos \theta = p$ 2) $x \cos \theta + y \sin \theta = p$
 3) $x \sin \theta + y \cos \theta = p$ 4) $x \cos \theta - y \sin \theta = p$
11. The parametric equation of a line is given by

$$x = -2 + \frac{r}{\sqrt{10}} \text{ and } y = 1 + 3 \frac{r}{\sqrt{10}}$$

Then, for the line

- 1) intercept on the x-axis = $\frac{7}{3}$
 2) intercept on the y-axis = -7
 3) slope of the line = $1/3$ 4) slope of the line = 3

PROBLEMS ON DISTANCES

12. A straight line through the origin 'O' meets the parallel lines $4x + 2y = 9$ and $2x + y + 6 = 0$ at points P and Q respectively, then point O divides the segment PQ in the ratio
 1) 1 : 2 2) 3 : 4 3) 2 : 1 4) 4 : 3
13. The lengths of the perpendiculars from $(m^2, 2m)$, $(mn, m + n)$ and $(n^2, 2n)$ to the straight line $x \cos \alpha + y \sin \alpha + \sin \alpha \tan \alpha = 0$ are in
 1) A.P. 2) G.P. 3) H.P. 4) A.G.P.

14. The distance between the Straight lines $y = mx + c_1$, $y = mx + c_2$ is $|c_1 - c_2|$ then $m =$
 1) 0 2) 1 3) 2 4) 3
15. Distance between parallel lines $4x + 6y + 8 = 0$, $6x + 9y + 15 = 0$ is
 1) $2/\sqrt{13}$ 2) $1/\sqrt{13}$ 3) $3/\sqrt{13}$ 4) $4/\sqrt{13}$
16. $2x + 3y - 5 = 0$, $2x + 3y + 15 = 0$, $x + y - 7 = 0$, $x + y + 7 = 0$ are sides of a parallelogram. Then the centre of the parallelogram is
 1) $(-5, -5)$ 2) $(5, -5)$ 3) $(-5, 5)$ 4) $(5, 5)$
17. The distance of the point $(3, 5)$ from the line $2x + 3y - 14 = 0$ measured parallel to the line $x - 2y = 1$ is
 1) $\frac{7}{\sqrt{5}}$ 2) $\frac{7}{\sqrt{13}}$ 3) $\sqrt{5}$ 4) $\sqrt{13}$
18. Equation of the straight line passing through $(1, 1)$ and at a distance of 3 units from $(-2, 3)$ is
 1) $x - 2 = 0$ 2) $5x - 12y + 6 = 0$
 3) $5x - 12y + 7 = 0$ 4) $y - 1 = 0$

POSITION OF A POINT (S) W.R.TO LINE (S)

19. If L_1, L_2 denote the lines $x + 2y - 2 = 0$, $2x + 3y + 4 = 0$
 1) L_1 is nearer to origin than L_2
 2) L_2 is nearer to origin than L_1
 3) L_1, L_2 are equidistant from origin 4) can't say
20. If the point (a, a) falls between the lines $|x + y| = 2$, then:
 1) $|a| = 2$ 2) $|a| = 1$ 3) $|a| < 1$ 4) $|a| < \frac{1}{2}$
21. If $(2a - 3, a^2 - 1)$ is on the same side of the line $x + y - 4 = 0$ as that of origin then the set of values of 'a' is
 1) $(-4, 2)$ 2) $(-2, 4)$ 3) $(-7, 8)$ 4) $(-7, 5)$
22. The range of values of the ordinate of a point moving on the line $x = 1$ and always remaining in the interior of the triangle formed by the lines $y = x$, the x-axis and $x + y = 4$
 1) $(0, 1)$ 2) $(0, 2)$ 3) $(1, 2)$ 4) $(2, 1)$

POINT OF INTERSECTION OF LINES AND CONCURRENCY OF LINES

23. If the line $\frac{x-1}{2} = \frac{y-2}{3} = t$ intersects the line $x + y = 8$ then $t =$
 1) 1 2) 2 3) 3 4) 4

24. Equation of line which is equally inclined to the axis and passes through a common points of family of lines $4acx + y(ab + bc + ca - abc) + abc = 0$ (where $a, b, c > 0$ are in H.P.) is

1) $y - x = \frac{7}{4}$ 2) $y + x = \frac{7}{4}$
 3) $y - x = \frac{1}{4}$ 4) $y + x = \frac{3}{4}$

25. If a, b, c in GP then the line $a^2x + b^2y + ac = 0$ always passes through the fixed point

1) (0, 1) 2) (1, 0) 3) (0, -1) 4) (1, -1)

26. If $U \equiv x + y - 2 = 0$, $V \equiv 2x - 3y + 1 = 0$, the point of intersection of the lines $50U + 7V = 0$, $3U + 11V = 0$ is

1) (0, 0) 2) (1, 0) 3) (0, 1) 4) (1, 1)

27. The straight lines $x + 2y - 9 = 0$, $3x + 5y - 5 = 0$ and $ax + by - 1$ are concurrent if the straight line $22x - 35y - 1 = 0$ passes through the point

1) (a, b) 2) (b, a) 3) (-a, b) 4) (-a, -b)

28. The equation of the line passing through the point of intersection of the lines $2x + y = 5$ and $y = 3x - 5$ and which is at the minimum distance from the point (1, 2) is

1) $x + y = 3$ 2) $x - y = 1$ 3) $x - 2y = 0$ 4) $2x + 5y = 7$

29. Given a family of lines $a(2x + y + 4) + b(x - 2y - 3) = 0$. The number of lines belonging to the family at a distance $\sqrt{10}$ from $P(2, -3)$ is

1) 0 2) 1 3) 2 4) 4

ANGLE BETWEEN LINES

30. The acute angle between the lines $lx + my = l + m$, $l(x - y) + m(x + y) = 2m$ is

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{3}$

31. The angle between the lines $x \cos \alpha + y \sin \alpha = p_1$ and $x \cos \beta + y \sin \beta = p_2$ where $\alpha > \beta$ is

1) $\alpha + \beta$ 2) $\alpha - \beta$ 3) $\alpha\beta$ 4) $2\alpha - \beta$

32. One vertex of an equilateral triangle is (2, 3) and the equation of one side is $x - y + 5 = 0$. Then the equations to other sides are

1) $y - 3 = -(2 \pm \sqrt{3})(x - 2)$ 2) $y - 3 = (\sqrt{2} \pm 1)(x - 2)$

3) $y - 3 = (\sqrt{3} \pm 1)(x - 2)$ 4) $y - 3 = (\sqrt{5} \pm 1)(x - 2)$

33. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$. The slope of the third side is

1) $-3, \frac{1}{3}$ 2) $3, -\frac{1}{3}$ 3) $3, \frac{1}{3}$ 4) $-3, -\frac{1}{3}$

34. Let there are two lines $2x + 3y + \lambda = 0$ and $\lambda x - 3y - 1 = 0$. If the origin lies in the obtuse angle then

1) $\lambda = \frac{9}{2}$ 2) $-2 < \lambda < 0$

3) $0 < \lambda < \frac{9}{2}$ 4) None of these

TRIANGLES, AREA OF THE TRIANGLE

35. The area of the triangle formed by the axes and the line $(\cosh \alpha - \sinh \alpha)x + (\cosh \alpha + \sinh \alpha)y = 2$ in square units is

1) 4 2) 3 3) 2 4) 1

36. The equation to the base of an equilateral triangle is $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y + 2\sqrt{3} = 0$ and opposite vertex is A(1, 1) then the Area of the triangle is

1) $3\sqrt{2}$ 2) $3\sqrt{3}$ 3) $2\sqrt{3}$ 4) $4\sqrt{3}$

37. Equation of the line on which the perpendicular from the origin makes an angle of 30° with X-axis and which forms a triangle of area $\frac{50}{\sqrt{3}}$ with the axes is

1) $\sqrt{2}x + 2y = 9$ 2) $\sqrt{2}x + 3y = \pm 9$

3) $\sqrt{3}x - y = \pm 10$ 4) $\sqrt{3}x + y = \pm 10$

QUADRILATERALS AND AREA OF THE QUADRILATERALS

38. The point (2, 3) is reflected four times about co-ordinate axes continuously starting with x-axis. The area of quadrilateral formed in sq. units is

1) 24 2) 6 3) 12 4) 5

39. Area of the quadrilateral formed by the lines $4y - 3x - a = 0$, $3y - 4x + a = 0$, $4y - 3x - 3a = 0$, $3y - 4x + 2a = 0$ is

1) $\frac{a^2}{5}$ 2) $\frac{a^2}{7}$ 3) $\frac{2a^2}{7}$ 4) $\frac{2a^2}{9}$

40. Two sides of a rectangle are $3x + 4y + 5 = 0$, $4x - 3y + 15 = 0$ and its one vertex is (0, 0). Then the area of the rectangle is

1) 4 2) 3 3) 2 4) 1

41. The area enclosed with in the curve $|x| + |y| = 1$ is

1) 1 2) 2 3) $2\sqrt{2}$ 4) 4

FOOT AND IMAGE

42. Foot of the perpendicular of $(6, 8)$ in the line $x = y$ is
 1) $(6, 6)$ 2) $(7, 7)$ 3) $(-6, -6)$ 4) $(-7, -7)$
43. P is the midpoint of the part of the line $3x + y - 2 = 0$ intercepted between the axes. Then the image of P in origin is
 1) $\left(-1, -\frac{1}{3}\right)$ 2) $\left(-\frac{1}{3}, -4\right)$
 3) $\left(-\frac{1}{3}, -1\right)$ 4) $(-2, -3)$
44. The image of the point $P(3, 5)$ with respect to the line $y = x$ is the point Q and the image of Q with respect to the line $y = 0$ is the point $R(a, b)$, then $(a, b) =$
 1) $(5, 3)$ 2) $(5, -3)$ 3) $(-5, 3)$ 4) $(-5, -3)$
45. The equation of perpendicular bisector of AB and AC of a triangle ABC are $x - y - 5 = 0$ and $x + 2y = 0$ respectively. If $A = (1, -2)$ then the equation of $\overline{BC}$ is
 1) $14x + 2y - 41 = 0$ 2) $11x + 2y - 25 = 0$ 3) $2x - y - 10 = 0$ 4) $14x - 23y + 40 = 0$

CENTROID, CIRCUMCENTRE, ORTHOCENTRE AND INCENTRE

46. Let $O(0, 0)$, $P(3, 4)$, $Q(6, 0)$ be the vertices of the triangle OPQ . The point R inside the triangles OPQ is such that the triangles OPR , PQR , OQR are of equal area. The coordinates of R are
 1) $\left(\frac{4}{3}, 3\right)$ 2) $\left(3, \frac{2}{3}\right)$ 3) $\left(3, \frac{4}{3}\right)$ 4) $\left(\frac{4}{3}, \frac{2}{3}\right)$
47. If the circumcentre of the triangle lies at $(0, 0)$ and centroid is mid point of the line joining the points $(2, 3)$ and $(4, 7)$, then its orthocentre lies on the line
 1) $5x - 3y = 0$ 2) $5x - 3y + 6 = 0$
 3) $5x + 3y = 0$ 4) $5x + 3y + 6 = 0$
48. The orthocentre of the triangle formed by the lines $x + y = 6$, $2x + y = 4$ and $x + 2y = 5$ is
 1) $(10, -11)$ 2) $(-10, 11)$
 3) $(11, -10)$ 4) $(-11, -10)$

49. The equation $x + 2y = 3$ represents the side BC of $\triangle ABC$; where co-ordinates of A are $(1, 2)$. If the x -coordinate of the orthocentre of $\triangle ABC$ is 3 then the y -coordinates of the orthocentre is :
 1) 4 2) 6 3) 8 4) 10
50. The vertices A, B of a triangle are $(2, 5)$, $(4, -11)$. If C moves on the line $L \equiv 9x + 7y + 4 = 0$, then the locus of centroid of triangle ABC is parallel to
 1) AB 2) AC 3) BC 4) L
51. Two sides of a triangle are $y = m_1 x$ and $y = m_2 x$; m_1, m_2 are the roots of the equation $x^2 + ax - 1 = 0$. For all values of 'a' the orthocentre of the triangle lies at
 1) $(1, 1)$ 2) $(2, 2)$ 3) $\left(\frac{3}{2}, \frac{3}{2}\right)$ 4) $(0, 0)$

ANGULAR BISECTORS

52. Equation of the line equidistant from the lines $2x + y + 4 = 0$, $3x + 6y - 5 = 0$ is
 1) $3x - 3y + 17 = 0$ 2) $5x + 7y - 5 = 0$
 3) $3x - 3y + 19 = 0$ 4) $9x - 9y + 17 = 0$
53. Find the equation of the bisector of the angle between the lines $x + 2y - 11 = 0$, $3x - 6y - 5 = 0$ which contains the point $(1, -3)$.
 1) $2x - 19 = 0$ 2) $2x + 19 = 0$
 3) $3x - 19 = 0$ 4) $3x + 19 = 0$
54. The line $3x - 3y + 17 = 0$ bisects the angle between a pair of lines of which one line is $2x + y + 4 = 0$, then the equation to the other line is
 1) $3x + 6y - 5 = 0$ 2) $3x + 6y - 7 = 0$
 3) $7x + 14y = 0$ 4) $4x - y + 3 = 0$
55. The equation of a straight line passing through the point $(4, 5)$ and equally inclined to the lines $3x = 4y + 7$ and $5y = 12x + 6$ is
 1) $9x - 7y = 1$ 2) $9x + 7y = 1$
 3) $7x - 9y = 1$ 4) $7x - 9y = 17$
56. If $2x + y - 4 = 0$ is bisector of the angle between the lines $a(x - 1) + b(y - 2) = 0$, $c(x - 1) + d(y - 2) = 0$, then the other bisector is
 1. $x - 2y + 1 = 0$ 2. $x - 2y - 3 = 0$
 3. $x - 2y + 3 = 0$ 4. $x - 2y - 5 = 0$

OPTIMIZATION AND REFLECTION IN SURFACE

57. A ray of light passing through the point $(8, 3)$ and is reflected at $(14, 0)$ on x axis. Then the equation of the reflected ray
 1) $x + y = 14$ 2) $x - y = 14$ 3) $2y = x - 14$ 4) $3y = x - 14$

58. Let P(1,1) and Q(3,2) be given points. The point R on the x-axis such that PR+RQ is minimum is

- 1) $\left(\frac{5}{3}, 0\right)$ 2) (2,0) 3) (3,0) 4) $\left(\frac{3}{2}, 0\right)$

MISCELLANEOUS PROBLEMS

59. The vertices of triangle are

$A(m, n)$, $B(12, 19)$ and $C(23, 20)$, where m and n are integer. If its area is 70 and the slope of the median through A is -5, then m+n is

- 1) 47 2) 27 3) 107 4) 43

60. Number of circles touching the lines $3x+4y-1=0$, $4x-5y+2=0$ and $6x+8y+3=0$ is
1) 0 2) 2 3) 4 4) infinite

61. The point on the line $3x+4y=5$ which is equidistance from (1,2) and (3,4) is
(EAM 2009)

- 1) (7, -4) 2) (15, -10) 3) $\left(\frac{1}{7}, \frac{8}{7}\right)$ 4) $\left(0, \frac{5}{4}\right)$

62. The number of points p(x,y) with natural numbers as coordinates that lie inside the quadrilateral formed by the lines $2x+y=2$, $x=0$, $y=0$ and $x+y=5$ is
(EAM- 2011)

- 1) 12 2) 10 3) 6 4) 4

63. A point moves in the xy plane such that the sum of its distance from two mutually perpendicular lines is always equal to 5 units. The area (in square units) enclosed by the locus of the point (EAM- 2012)

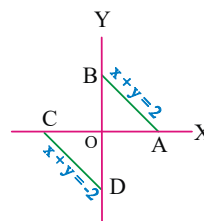
- 1) $\frac{25}{4}$ 2) 25 3) 50 4) 100

LEVEL-II (H.W)- KEY

- 1) 4 2) 4 3) 1 4) 1 5) 2 6) 1
7) 4 8) 4 9) 4 10) 1 11) 4 12) 2
13) 2 14) 1 15) 2 16) 2 17) 3 18) 3
19) 1 20) 3 21) 1 22) 1 23) 1 24) 1
25) 3 26) 4 27) 2 28) 1 29) 2 30) 1
31) 2 32) 1 33) 1 34) 3 35) 3 36) 3
37) 4 38) 1 39) 3 40) 2 41) 2 42) 2
43) 3 44) 2 45) 2 46) 3 47) 1 48) 4
49) 2 50) 4 51) 4 52) 1 53) 3 54) 1
55) 1 56) 3 57) 3 58) 1 59) 3 60) 2
61) 2 62) 3 63) 3

LEVEL-II (H.W)-HINTS

- $\tan \frac{3\pi}{4} = \frac{2-k}{1-k}$
- Slope of the parallel lines are equal
- $A(-6, 0)B(0, -4)$ perpendicular bisector of AB
is $y+2 = \frac{3}{2}(x+3)$
- Equation of the line is $y(t_1+t_2) - 2at_1(t_1+t_2) = 2x - 2at_1^2$
- by solving AB, BD we get $B(-5/3, 4/3)$
by solving AD, BD we get $D(2/3, -7/3)$
mid point of B, D lies on AC
- $a=8, b=6$
- sum of the intercepts of the line $2x+3y+k=0$ is 15.
 $\frac{-k}{2} + \frac{-k}{3} = 15$
- $\left(\frac{a}{2}, \frac{b}{2}\right) = (1, 2)$ 9. $\frac{nx}{x_1} + \frac{my}{y_1} = m+n$
- The line makes an angle θ with x-axis then its perpendicular makes an angle $\alpha = \theta - 90^\circ$
- Point $(-2, 1)$ slope = 3
- $\therefore \left| \frac{OP}{OQ} \right| = \left| \frac{9/\sqrt{20}}{-6/\sqrt{5}} \right| = \left| \frac{9}{12} \right| = \frac{3}{4} = 3:4$
- Put $\alpha = 0$ then given equation of line is $x=0$
- $\frac{|c_1 - c_2|}{\sqrt{1+m^2}} = |c_1 - c_2|$
- multiply first equation by 3 and second equation by 2
- Centre lies on a line parallel to given lines and mid way between them
- $\left| \frac{ax_1 + by_1 + c}{a \cos \theta + b \sin \theta} \right|$ where $\tan \theta = \frac{1}{2}$
- verification 19. verification
- From the figure $-1 < a < 1$ i.e. $|a| < 1$.



THE STRAIGHT LINE

21. since $L_0 < 0$ we have $L_A < 0$
22. From figure $0 < \text{ordinate of } P < 1$
23. Point $(2t+1, 3t+2)$ lies on $x+y=8$
24. Lines can be written
- $$\frac{4}{b}x + y\left(\frac{3}{b}\right) + 1 - y = 0, \frac{1}{b}(4x + 3y) + 1 - y = 0$$
- $$\Rightarrow \text{Lines are concurrent at } \left(-\frac{3}{4}, 1\right)$$
- $$\therefore \text{Required line is } y - 1 = \pm 1\left(x + \frac{3}{4}\right)$$
25. Given equation is $a^2x + b^2(y+1) = 0$
26. Point of intersection of u & v
- $$\begin{vmatrix} 1 & 2 & -9 \\ 3 & 5 & -5 \\ a & b & -1 \end{vmatrix} = 0$$
- 27.
28. The point $(1, 2)$ lies on $L_1 + \lambda L_2 = 0$
29. $P = \frac{a(4-3+4) + b(2+6-3)}{\sqrt{(2a+b)^2 + (a-2b)^2}} = \sqrt{10}$
- $$\begin{aligned} 25(a+b)^2 &= 10(5a^2 + 5b^2) \\ 25(a-b)^2 &= 0; a = b \end{aligned}$$
30. $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
31. $\cos \theta = \frac{a_1 a_2 + b_1 b_2}{\sqrt{a_1^2 + b_1^2} \sqrt{a_2^2 + b_2^2}}$
32. apply $\frac{m \pm \tan \alpha}{1 \mp m \tan \alpha}$
33. Apply $\frac{m_1 - m}{1 + m_1 m} = -\frac{m_2 - m}{1 + m_2 m}$
34. Origin will lie in obtuse angle if $cc'(aa' + bb') > 0$
- $$\Rightarrow -\lambda(2\lambda - 3 \times 3) > 0 \Rightarrow -\lambda(2\lambda - 9) > 0$$
- $$\Rightarrow 9\lambda - 2\lambda^2 > 0 \quad \lambda \in \left(0, \frac{9}{2}\right)$$
35. $\Delta = \frac{c^2}{2|ab|}$
36. Area of an equilateral triangle is
- $$\frac{h^2}{\sqrt{3}} \text{ where } h \text{ is the height of the triangle}$$

JEE-MAIN-JR-MATHS VOL-III

37. Equation of the line is $x \cos 30^\circ + y \sin 30^\circ = P$
38. $A(2, 3) B(-2, 3) C(-2, -3) D(2, -3)$
39. The area of the parallelogram formed by the lines $a_1x + b_1y + c_1 = 0, a_2x + b_2y + d_1 = 0, a_1x + b_1y + c_2 = 0, a_2x + b_2y + d_2 = 0$ is $\left| \frac{(c_1 - c_2)(d_1 - d_2)}{a_1 b_2 - a_2 b_1} \right|$ Sq. units.
40. The perpendicular distance from origin lines are a, b. Then area = ab
41. From the diagram required area = $4(\text{area of } \Delta) = 2$
42. Use foot of the perpendicular formula
43. Midpoint of intercepts is $\left(\frac{1}{3}, 1\right)$
44. Image of $P(3, 5)$ w.r to the line $y = x$ is $Q(5, 3)$
45. B, C are images of A w.r. to given lines
46. Centroid ΔOPQ
47. circumcentre, centroid and orthocentre lies on same line.
48. Use formula given in synopsis
49. $AD \perp BC$
- $$\Rightarrow \left[\frac{\lambda - 2}{3 - 1} \right] \left[-\frac{1}{2} \right] = -1$$
- or $2 - \lambda = -4$
- $$-\lambda = -6 \Rightarrow \lambda = 6.$$
50. Choose $C = (\alpha, \beta)$
- $$G(x_1, y_1) = \left(\frac{\alpha + 6}{3}, \frac{\beta - 6}{3} \right)$$
- $$\Rightarrow \alpha = 3x_1 - 6, \beta = 3y_1 + 6$$
- Substance (α, β) lies on $L=0$
51. given triangle is right angled at origin
52. Angular bisector
53. Using formula given in synopsis
54. Verify angular bisector formula
55. $(x_1, y_1) = (4, 5)$
- Slope of angle of bisector are $\frac{9}{7}, \frac{-7}{9}$
56. required bisector is perpendicular to given and passes through $(1, 2)$

57. Write the image of $(8,3)$ in X-axis and write the equation through that point and $(14,0)$
58. Image of P in x-axis is $P^1 = (1, -1)$, R is intersection of x-axis and line QP^1
59. $Area = 70 \Rightarrow -m + 11n = 337$
 $median\ slope = -5 \Rightarrow 5m + n = 107$
60. Two lines are parallel
61. Required point $p(x_1, y_1)$ lies on given line
 then $3x_1 + 4y_1 = 5$ and $PA=PB \Rightarrow x_1 + y_1 = 5$
62. Draw the diagram and observe the points $(1,1), (1,2), (2,1), (2,2), (1,3)$ and $(3,1)$ are inside of quadrilateral.
63. From given data $|x| + |y| = 5$ hence required

$$area = \frac{2(5)^2}{|(1)(1)|} = 50$$



LEVEL - III

1. The line $3x-2y = 24$ meets x-axis at A and y-axis at B. The perpendicular bisector of AB meets the line through $(0, -1)$ and parallel to x-axis at C. Then C is
- 1) $\left(\frac{-7}{2}, -1\right)$ 2) $\left(\frac{-15}{2}, -1\right)$
 3) $\left(\frac{-11}{2}, -1\right)$ 4) $\left(\frac{-13}{2}, -1\right)$
2. A square of side "a" lies above the x-axis and has one vertex at the origin. The side passing through the origin makes an angle α where $0 < \alpha < \frac{\pi}{4}$ with the positive direction of x - axis. The equation of its diagonal not passing through the origin is
- 1) $y(\cos \alpha + \sin \alpha) + x(\cos \alpha - \sin \alpha) = a$
 2) $y(\cos \alpha - \sin \alpha) - x(\sin \alpha - \cos \alpha) = a$
 3) $y(\cos \alpha + \sin \alpha) + x(\sin \alpha - \cos \alpha) = a$
 4) $y(\cos \alpha + \sin \alpha) + x(\cos \alpha + \sin \alpha) = a$

3. A particle is moving in a straight line and at some moment it occupied the positions $(5,2)$ and $(-1,-2)$. Then the position of the particle when it is on x-axis is
 1) $(-2, 0)$ 2) $(0, 2)$ 3) $(2, 0)$ 4) $(4, 0)$
4. If PS is the median of the triangle with vertices $P(2,2)$, $Q(6,-1)$ and $R(7,3)$ then the equation of the line passing through $(1,-1)$ and parallel to $\overline{PS}$ (AIEEE 2014)
 1) $4x-7y-11=0$ 2) $2x+9y+7=0$
 3) $4x+7y+3=0$ 4) $2x-9y-11=0$
5. The Point $P(2, 1)$ is shifted by $3\sqrt{2}$ parallel to the line $x + y = 1$, in the direction of increasing ordinate, to reach Q. The image of Q by the line $x + y = 1$ is
 1) $(5, -2)$ 2) $(-1, 4)$ 3) $(3, -4)$ 4) $(-3, 2)$
6. Distance of origin from line $(1 + \sqrt{3})y + (1 - \sqrt{3})x = 10$ along the line $y = \sqrt{3}x + k$ is
 1) $5/\sqrt{2}$ 2) $5/\sqrt{2} + k$ 3) 10 4) 5
7. One of the diagonals of a square is the portion of the line $\frac{x}{2} + \frac{y}{3} = 2$ intercepted between the axes. Then the extremities of the other diagonal are
 1) $(5,5), (-1,1)$ 2) $(0,0), (4,6)$
 3) $(0,0), (-1,1)$ 4) $(5,5), (4,6)$
8. If the line $y = \sqrt{3}x$ cuts the curve $x^3 + y^3 + 3xy + 5x^2 + 3y^2 + 4x + 5y - 1 = 0$ at the points A, B, C then OA.OB.OC is
 1) $\frac{4}{13}(3\sqrt{3}-1)$ 2) $3\sqrt{3}+1$
 3) $\frac{2}{\sqrt{3}}+7$ 4) $3\sqrt{3}-1$
9. Each side of a square is of length 4. The centre of the square is $(3, 7)$ and one of its diagonals is parallel to $y=x$. Then co-ordinates of its vertices are
 1) $(1,5), (1,9), (5,9), (5,5)$
 2) $(2,5), (2,7), (4,7), (4,4)$
 3) $(2,5), (2,6), (3,5), (3,6)$
 4) $(5,2), (6,2), (5,3), (6,3)$

10. If the line $y - \sqrt{3}x + 3 = 0$ cuts the curve $y^2 = x + 2$ at A and B and point on the line P is $(\sqrt{3}, 0)$ then $|PA \cdot PB| =$

1) $\frac{4(\sqrt{3} + 2)}{3}$ 2) $\frac{4(2 - \sqrt{3})}{3}$
3) $\frac{4\sqrt{3}}{2}$ 4) $\frac{2(\sqrt{3} + 2)}{3}$

11. Points A and B are in the first quadrant; point 'O' is the origin. If the slope of OA is 1, slope of OB is 7 and $OA = OB$, then the slope of AB is

1) $-\frac{1}{5}$ 2) $-\frac{1}{4}$ 3) $-\frac{1}{3}$ 4) $-\frac{1}{2}$

12. The line joining the points A(3,0) and B(5,2) is rotated about A in the anticlockwise direction through an angle of 15° . If B goes to C in the new position now the line joining A and C is rotated about A in the anticlockwise direction through an angle of 45° of C goes to D in the new position, then the coordinates of D are

1) $(4 - \sqrt{3}, \sqrt{3} - 1)$ 2) $(4 + \sqrt{3}, \sqrt{3} - 1)$
3) $(4 - \sqrt{3}, \sqrt{3} + 1)$ 4) $(4 + \sqrt{3}, \sqrt{3} + 1)$

13. The equation of the line passing through (1,2) and having a distance equal to 7 units from the point (8,9) is

1) $y = 3x - 1$ 2) $y = 2$
3) $x = 1$ 4) $x + y = 3$

14. Find the values of non-negative real numbers $h_1, h_2, h_3, k_1, k_2, k_3$ such that the algebraic sum of the perpendiculars drawn from points $(2, k_1), (3, k_2), (7, k_3), (h_1, 4), (h_2, 5), (h_3, -3)$ on a variable line passing through $(2, 1)$ is zero.

1) $h_1 = h_2 = h_3 = k_1 = k_2 = k_3 = 0$
2) $h_1 = h_2 = h_3 = k_1 = k_2 = k_3 = 1$
3) $h_1 = h_2 = h_3 = k_1 = k_2 = k_3 = 2$
4) $h_1 = h_2 = h_3 = k_1 = k_2 = k_3 = 4$

15. Three lines $x + 2y + 3 = 0$; $x + 2y - 7 = 0$ and $2x - y - 4 = 0$ form the three sides of two squares. The equation to the fourth side of each square is

1) $2x - y + 14 = 0$ and $2x - y + 6 = 0$
2) $2x - y + 14 = 0$ and $2x - y - 6 = 0$
3) $2x - y - 14 = 0$ and $2x - y - 6 = 0$
4) $2x - y - 14 = 0$ and $2x - y + 6 = 0$

16. The line L given by $\frac{x}{5} + \frac{y}{b} = 1$ passes through the point (13, 32). The line K is parallel to L and has the equation $\frac{x}{c} + \frac{y}{3} = 1$. Then the distance between L and K is

[AIEEE-2010]

1) $\sqrt{17}$ 2) $\frac{17}{\sqrt{15}}$ 3) $\frac{23}{\sqrt{17}}$ 4) $\frac{23}{\sqrt{15}}$

17. If the line $2x + y = k$ passes through the point which divides the line segment joining the points (1, 1) and (2, 4) in the ratio 3 : 2, then k equals [AIEEE - 2012]

1) $\frac{29}{5}$ 2) 5 3) 6 4) $\frac{11}{5}$

18. If the point $P(a^2, a)$ lies in the region corresponding to the acute angle between the lines $2y = x$ and $4y = x$, then

1) $a \in [2, 4]$ 2) $a \in (2, 4)$
3) $a \in [2, 5)$ 4) $a \in (2, 4)$

19. The set of values of 'b' for which the origin and the point (1,1) lie on the same side of the straight line

$a^2x + aby + 1 = 0, \forall a \in R, b > 0$ are

1) $b \in (2, 4)$ 2) $b \in (0, 2)$
3) $b \in [0, 2]$ 4) $b \in [0, 3]$

20. The equations of sides of a triangle are $7x - 5y - 11 = 0$, $8x + 3y + 31 = 0$, $x + 8y - 19 = 0$. Then the point (0,0) lies

1) inside of triangle 2) outside of triangle
3) on the triangle 4) can't say

21. If the points of intersection of lines

$$L_1 : y - m_1x - k = 0 \text{ and}$$

$L_2 : y - m_2x - k = 0$ ($m_1 \neq m_2$) lies inside the triangle formed by the lines

$$2x + 3y = 1, x + 2y = 3 \text{ and } 5x - 6y - 1 = 0,$$

then true set of values of k are

$$1) \left(\frac{1}{3}, \frac{3}{2}\right) \quad 2) \left(\frac{1}{2}, 1\right) \quad 3) \left(0, \frac{3}{2}\right) \quad 4) \left(\frac{-3}{2}, 0\right)$$

22. The range of value of α such that $(0, \alpha)$ lies on or inside the triangle formed by the lines $y + 3x + 2 = 0$, $3y - 2x - 5 = 0$, $4y + x - 14 = 0$ is

$$1) 5 < \alpha \leq 7 \quad 2) \frac{1}{2} \leq \alpha \leq 1 \quad 3) \frac{5}{3} \leq \alpha \leq \frac{7}{2} \quad 4) \frac{1}{3} \leq \alpha \leq \frac{1}{2}$$

23. The lines $x + y = |a|$, and $ax - y = 1$ intersect each other in the first quadrant then the set of all possible values of a is the interval

[AIEEE-2011]

$$1) (0, \infty) \quad 2) (1, \infty) \quad 3) (-1, \infty) \quad 4) [-1, 1]$$

24. Let a, b, c and d be non zero numbers. If the point of intersection of the lines

$$4ax + 2ay + c = 0 \text{ and } 5bx + 2by + d = 0 \text{ lies in the fourth quadrant and is equidistance from the two axes then (MAINS 2014)}$$

$$1) 2bc - 3ad = 0 \quad 2) 2bc + 3ad = 0$$

$$3) 3bc - 2ad = 0 \quad 4) 3bc + 2ad = 0$$

25. If the lines $x + ay + a = 0$, $bx + y + b = 0$, $cx + cy + 1 = 0$ (a, b, c being distinct and $\neq 1$) are concurrent

$$\text{then the value of } \frac{a}{a-1} + \frac{b}{b-1} + \frac{c}{c-1} =$$

$$1) -1 \quad 2) 0 \quad 3) 1 \quad 4) 3$$

26. If $4a^2 + 9b^2 - c^2 + 12ab = 0$ then the family of straight lines $ax + by + c = 0$ is concurrent at

$$1) (2, 3) \text{ or } (-2, -3) \quad 2) (2, -3) \text{ or } (-2, 6)$$

$$3) (-2, -4) \text{ or } (-2, 3) \quad 4) (2, 5) \text{ or } (-1, -5)$$

27. If $a^2 - b^2 - c^2 - 2bc = 0$, then the family of lines $ax + by + c = 0$ are concurrent at the points

$$1) (1, -1) \quad 2) (-1, 1) \quad 3) (1, 0) \quad 4) (-1, -1)$$

28. If $t_1 \neq t_2 \neq t_3$ and the lines $t_1x + y = 2at_1 + at_1^3$; $t_2x + y = 2at_2 + at_2^3$; $t_3x + y = 2at_3 + at_3^3$ are concurrent then $t_1 + t_2 + t_3$ is

$$1) 0 \quad 2) -1 \quad 3) 1 \quad 4) 2$$

29. Consider the family of lines

$$(x + y - 1) + \lambda(2x + 3y - 5) = 0 \text{ and}$$

$$(3x + 2y - 4) + \mu(x + 2y - 6) = 0,$$

equation of a straight line that belongs to both the families is

$$1) x - 2y - 8 = 0 \quad 2) x - 2y + 8 = 0$$

$$3) 2x + y - 8 = 0 \quad 4) 2x - y - 8 = 0$$

30. If a, b and c are three consecutive odd integers then the variable line $ax + by + c = 0$ always passes through

$$1) (2, 1) \quad 2) (1, 2) \quad 3) (-1, 2) \quad 4) (1, -2)$$

31. One vertex of an equilateral triangle is $(2, 3)$ and the equation of one side is $x - y + 5 = 0$ then the equations to the other sides are

$$1) y - 3 = -(2 \pm \sqrt{3})(x - 2)$$

$$2) y - 3 = (\sqrt{2} \pm 1)(x - 2)$$

$$3) y - 3 = (\sqrt{3} \pm 1)(x - 2)$$

$$4) y - 3 = (\sqrt{5} \pm 1)(x - 2)$$

32. Let $P(2, -4)$ and $Q(3, 1)$ be two given points. Let $R(x, y)$ be a point such that $(x - 2)(x - 3) +$

$$(y - 1)(y + 4) = 0. \text{ If area of } \Delta PQR \text{ is } \frac{13}{2}, \text{ then}$$

the number of possible positions of R are

$$1) 2 \quad 2) 3 \quad 3) 4 \quad 4) 6$$

33. If the base of an isosceles triangle is of length $2p$ and the length of the altitude dropped to the base is q , then the distance from the mid point of the base to the side of the triangle is

$$1) \frac{pq}{\sqrt{p^2 + q^2}} \quad 2) \frac{2pq}{\sqrt{p^2 + q^2}}$$

$$3) \frac{3pq}{\sqrt{p^2 + q^2}} \quad 4) \frac{4pq}{\sqrt{p^2 + q^2}}$$

34. If m_1 and m_2 are the roots of the equation

$$x^2 - ax - a - 1 = 0, \text{ then the area of the triangle formed by the three straight lines}$$

$$y = m_1x, y = m_2x \text{ and } y = a(a \neq -1) \text{ is}$$

$$1) \frac{-a^2(a+2)}{2(a+1)} \text{ if } a > -1 \quad 2) \frac{+a^2(a+2)}{2(a+1)} \text{ if } a < -1$$

$$3) \frac{-a^2(a+2)}{2(a+1)} \text{ if } -2 < a < -1 \quad 4) \frac{a^2(a+2)}{2(a+1)} \text{ if } a > -2$$

35. The equation of a straight line L is $x+y=2$, and L_1 is another straight line perpendicular to L and passes through the point $(1/2, 0)$, then area of the triangle formed by the y -axis and the lines L, L_1 is
 1) $\frac{25}{8}$ 2) $\frac{25}{16}$ 3) $\frac{25}{4}$ 4) $\frac{25}{12}$
36. In an isosceles triangle OAB , O is the origin and $OA=OB=6$. The equation of the side AB is $x-y+1=0$. Then the area of the triangle is
 1) $2\sqrt{21}$ 2) $\sqrt{142}$ 3) $\frac{\sqrt{142}}{2}$ 4) $\frac{\sqrt{71}}{2}$
37. An equilateral triangle is constructed between two parallel lines $\sqrt{3}x+y-6=0$ and $\sqrt{3}x+y+9=0$ with base on one and vertex on the other. Then the area of triangle is
 1) $\frac{200}{\sqrt{3}}$ 2) $\frac{225}{4\sqrt{3}}$ 3) $\frac{225}{\sqrt{3}}$ 4) $\frac{200}{4\sqrt{3}}$
38. Area of triangle formed by the lines $2x+y-3=0$, $x+4y-5=0$ and $3x+5y-1=0$ is
 1) $15/2$ 2) $49/2$ 3) $27/56$ 4) $7/2$
39. If $f(x+y)=f(x)f(y)$ for all x and y if $f(1)=2$, then area enclosed by $3|x|+2|y|\leq 8$ is
 1) $f(5)sq.units$ 2) $f(6)sq.units$ 3) $1/3f(6)sq.units$ 4) $f(4)sq.units$
40. Four sides of a quadrilateral are given by the equation $xy(x-2)(y-3)=0$, then the equation of the line parallel to $x-4y=0$ that divides the quadrilateral into two equal parts is
 1) $x-4y+5=0$ 2) $x-4y-5=0$
 3) $x-4y+1=0$ 4) $x-4y-1=0$
41. L_1 and L_2 are two intersecting lines and the angle between the image of L_1 w.r.t L_2 and that of L_2 w.r.t L_1 is 45° . Then the angle between L_1 and L_2 is
 1) 20° 2) 15° 3) 45° 4) 60°
42. L_1 and L_2 are two intersecting lines. If the image of L_1 w.r.t L_2 and that of L_2 w.r.t L_1 coincide, then the angle between L_1 and L_2 is
 1) 35° 2) 60° 3) 90° 4) 45°
43. For all values of θ all the lines represented by the equation $(2\cos\theta+3\sin\theta)x+(3\cos\theta-5\sin\theta)y-(5\cos\theta-2\sin\theta)=0$ passes through a fixed point then the reflection of that point with respect to the line $x+y=\sqrt{2}$ is
 1) $(\sqrt{2}+1, \sqrt{2}+1)$ 2) $(\sqrt{2}-1, \sqrt{2}-1)$
 3) $(\sqrt{3}-1, \sqrt{3}-1)$ 4) $(\sqrt{3}+1, \sqrt{3}+1)$
44. The combined equation of straight lines h at t can be obtained by reflecting the lines $y=|x-2|$ in the y -axis is
 1) $y^2+x^2+4x+4=0$
 2) $y^2+x^2-4x+4=0$
 3) $y^2-x^2+4x-4=0$
 4) $y^2-x^2-4x-4=0$
45. In $\triangle ABC$, $B=(0, 0)$, $AB=2$, $\angle ABC = \frac{\pi}{3}$ and the middle point of BC has the co-ordinates $(2, 0)$. Then the centroid of triangle is
 1) $(\frac{5}{3}, \frac{1}{3})$ 2) $(\frac{5}{3}, \frac{1}{\sqrt{3}})$
 3) $(\frac{5}{\sqrt{3}}, \frac{1}{3})$ 4) $(\frac{5}{\sqrt{3}}, \frac{1}{\sqrt{3}})$
46. In triangle ABC , co-ordinates of A are $(-1, 3)$ and equation of medians and altitude through point B are $2x+y=8$ and $2x+3y=8$ respectively, then
 1) coordinates of C are $(4, 0)$
 2) coordinates of C are $(3, 9)$
 3) coordinates of C are $(3, 3)$
 4) coordinates of centroid are $(2, 2)$
47. The sides of a triangle are $x+y=1$, $7y=x$ and $\sqrt{3}y+x=0$. Then the following is an interior point of the triangle
 1) Circumcentre 2) Centroid
 3) Orthocentre 4) Cannot say
48. If the equations of the sides of a triangle are $2x+y=2$, $y=x$, $\sqrt{3}y+x=0$ then which of the following is an exterior point of triangle.
 1) orthocentre 2) incentre
 3) centroid 4) Cannot say

49. One vertex of the equilateral triangle with centroid at the origin and one side as $x+y-2=0$ is

1) $(-2, -2)$ 2) $(2, 2)$ 3) $(2, -2)$ 4) $(-2, 2)$

50. A ray of light is sent along the line $x-2y-3=0$. On reaching the line $3x-2y-5=0$, the ray is reflected from it. The equation of the line containing the the reflected ray.

1) $29x-2y+31=0$ 2) $29x+2y-31=0$
3) $29x-2y-31=0$ 4) $29x+2y+31=0$

51. A light ray coming along the line $3x+4y=5$ gets reflected from the line $ax+by=1$ and goes along the line $5x-2y=10$. Then,

1) $a = \frac{64}{115}, b = \frac{112}{15}$ 2) $a = \frac{14}{15}, b = \frac{-8}{115}$

3) $a = \frac{64}{115}, b = \frac{-8}{115}$ 4) $a = \frac{64}{15}, b = \frac{14}{15}$

52. If x_1, y_1 are roots of $x^2+8x-20=0, x_2, y_2$ are the roots of $4x^2+32x-57=0$ and x_3, y_3 are the roots of $9x^2+72x-112=0$, then the points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) where

$x_i < y_i$ for $i=1,2,3$

- 1) are collinear
2) form an equilateral triangle
3) form a right angled isosceles triangle
4) are concyclic

53. Triangle is formed by the coordinates $(0, 0), (0, 21)$ and $(21, 0)$. The number of integral coordinates strictly inside triangle (integral coordinates has both x and y as integers) :

1) 190 2) 105 3) 231 4) 205

54. Origin is the centre of the square with one of its vertices at $(3,4)$ then the other vertices are

1) $(-3, 4), (-3, -4), (3, -4)$
2) $(-4, 3), (-3, -4), (4, -3)$
3) $(-4, 3), (-4, -3), (3, -4)$
4) $(3, 4), (-4, -3), (4, -3)$

55. One side of a rectangle lies along the line $4x+7y+5=0$. Two vertices are $(-3,1), (1,1)$ then the remaining vertices are

1) $\left(\frac{1}{65}, \frac{-47}{65}\right), \left(\frac{-131}{65}, \frac{177}{65}\right)$

2) $\left(\frac{-1}{65}, \frac{47}{65}\right), \left(\frac{-131}{65}, \frac{177}{65}\right)$

3) $\left(\frac{1}{65}, \frac{-47}{65}\right), \left(\frac{131}{65}, \frac{-177}{65}\right)$

4) $(1, -47), (131, 47)$

56. All points lying inside the triangle formed by the points $(1,3), (5,0), (-1,2)$ satisfy

1) $2x+y-13=0$ 2) $3x+2y \geq 0$

3) $3x-4y-12 \leq 0$ 4) $4x+y=0$

57. If one vertex of an equilateral triangle of side a lies at the origin and the other lies on the line $x-\sqrt{3}y=0$, the coordinates of the third vertex are

1) $(0, -a)$

2) $(a, 0)$

3) $\left(\frac{a\sqrt{3}}{2}, \frac{a}{2}\right)$

4) $\left(\frac{a\sqrt{3}}{2}, -\frac{a}{2}\right)$

58. let AB be a line segment of length 4 units with the point A on the line $y=2x$ and B on the line $y=x$. Then the locus of middle point of all such line segment is

1) a parabola

2) an ellipse

3) a hyperbola

4) a circle

LEVEL-III - KEY

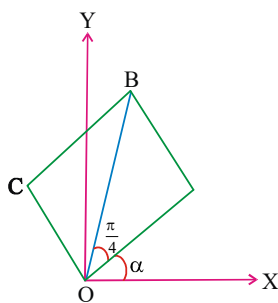
- 1) 1 2) 1 3) 3 4) 2 5) 4 6) 4
7) 1 8) 1 9) 1 10) 3 11) 4 12) 2
13) 2 14) 1 15) 4 16) 3 17) 3 18) 4
19) 2 20) 1 21) 1 22) 3 23) 2 24) 3
25) 3 26) 1 27) 2 28) 1 29) 2 30) 4
31) 1 32) 1 33) 1 34) 3 35) 2 36) 4
37) 2 38) 4 39) 3 40) 1 41) 2 42) 2
43) 2 44) 4 45) 2 46) 2 47) 2 48) 1
49) 1 50) 3 51) 3 52) 1 53) 1 54) 2
55) 1 56) 2 57) 4 58) 2

LEVEL-III - HINTS

1. perpendicular bisector of AB is

$2x+3y+10=0$(1)

$y=-1$(2), Solve the equations



2.

Slope of BC $\tan\left(\frac{\pi}{4} + \alpha\right)$

Slope of CA $-\cot\left(\frac{\pi}{4} + \alpha\right)$

Equation of CA

$$(y - a \sin \alpha) = -\cot\left(\frac{\pi}{4} + \alpha\right)(x - a \cos \alpha)$$

3. equation of line through given points is $2x - 3y - 4 = 0$ when cuts x-axis at $(2, 0)$

4. S = midpoint of Q, R = $(13/2, 1)$

slope of $\overline{PS} = -2/9$

5. $(x_1 + r \cos \theta, y_1 + r \sin \theta) = (-1, 4)$

$$r = 3\sqrt{2}, \theta = 3\pi/4$$

6. The slope of the line is $\sqrt{3}$

Find the equation of the line passing through origin. Then find point of intersection

7. The equation of the other diagonal is

$$\frac{x-2}{3} = \frac{y-3}{2} = r$$

$$\frac{r}{\sqrt{13}} = \frac{r\sqrt{3}}{\sqrt{13}}$$

For the extremities of the diagonal,

$$r = \pm\sqrt{13}. \text{ Hence } x-2 = \pm 3, y-3 = \pm 2$$

$$x = 5, -1 \text{ and } y = 5, 1$$

Therefore, the extremities of the diagonal are $(5, 5)$ and $(-1, 1)$.

8. $\therefore y = \sqrt{3}x = x \tan 60^\circ$

$$\therefore \frac{x-0}{\cos 60^\circ} = \frac{y-0}{\sin 60^\circ} = r; \therefore (x, y) = \left(\frac{r}{2}, \frac{r\sqrt{3}}{2}\right)$$

lies on

$$x^3 + y^3 + 3xy + 5x^2 + 3y^2 + 4x + 5y - 1 = 0$$

$$\therefore \frac{r^3}{8} + \frac{3\sqrt{3}r^3}{8} + \frac{3\sqrt{3}r^2}{4} + \frac{5r^2}{4} + \frac{9r^2}{4} + 2r + \frac{5\sqrt{3}r}{2} - 1 = 0$$

$$\Rightarrow \left(\frac{3\sqrt{3}+1}{8}\right)r^3 + \left(\frac{14+3\sqrt{3}}{4}\right)r^2 + \left(\frac{4+5\sqrt{3}}{2}\right)r - 1 = 0$$

$$\therefore OA \cdot OB \cdot OC = r_1 r_2 r_3 = \frac{1}{\left(\frac{3\sqrt{3}+1}{8}\right)}$$

9. Use $(x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$

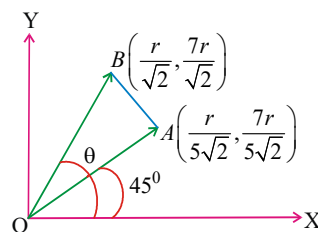
$$r = 2\sqrt{2} \text{ And } \theta = 45^\circ, 135^\circ$$

10. Let $PA = r$, then $A = \left(\sqrt{3} + \frac{r}{2}, \frac{\sqrt{3}r}{2}\right)$.

11. $\tan \theta = 7$

$$OA = OB = r$$

$$\sin \theta = \frac{7}{5\sqrt{2}}, \cos \theta = \frac{1}{5\sqrt{2}}$$

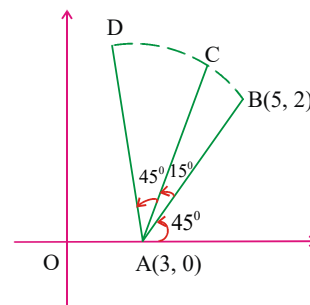


$$\text{Now, } m_{AB} = -1/2$$

12. Slope of $AB = 1$

$$r = AB = 2\sqrt{2}, \theta = 105^\circ$$

$$D = (x_1 + r \cos \theta, y_1 + r \sin \theta) = (4 + \sqrt{3}, \sqrt{3} - 1)$$



13. Distance from $(8, 9)$ to line $y - 2 = m(x - 1)$ is 7
 $\Rightarrow m = 0$

14. Let the equation of variable line be $ax + by + c = 0$,

it is given that $\sum_{i=1}^6 \frac{ax_i + by_i + c}{\sqrt{a^2 + b^2}} = 0$

$$\Rightarrow a \left(\frac{\sum x_i}{6} \right) + b \left(\frac{\sum y_i}{6} \right) + c = 0$$

So, the fixed point must be $\frac{\sum x_i}{6}, \frac{\sum y_i}{6}$. But fixed

point is $(2, 1)$ so $(2 + 3 + 7 + h_1 + h_2 + h_3) / 6 = 2$

$$\Rightarrow h_1 + h_2 + h_3 = 0; \Rightarrow h_1 = 0, h_2 = 0, h_3 = 0$$

(as h_1, h_2, h_3 are non-negative)

similarly, we get

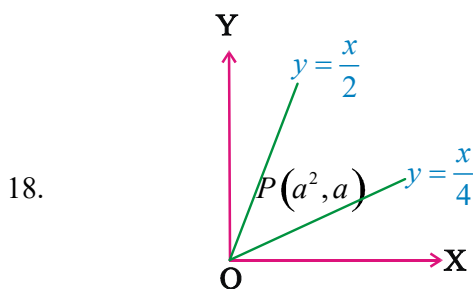
$$\frac{k_1 + k_2 + k_3 + 4 + 5 + 3}{6} = 1 \Rightarrow k_1 = k_2 = k_3 = 0$$

15. Hence line is $2x - y + \lambda = 0$

now $\left| \frac{\lambda + 4}{\sqrt{5}} \right| = \frac{10}{\sqrt{5}}; \lambda + 4 = \pm 10$

$$\lambda = 6 \text{ or } \lambda = -14 \Rightarrow (B)$$

16. Substitute the given point. Find 'b' and equate the slopes to find 'c' and apply distance between parallel lines.
17. Find ratio point and substitute in the line.



But $a^2 > 0$, hence point $P(a^2, a)$ lies in first quadrant.

We have $a - \left(\frac{a^2}{4} \right) > 0$ and $a - \left(\frac{a^2}{2} \right) < 0$

($\therefore (1, 0)$ and P lies on same side of $x - 2y = 0$

and $(1, 0)$ and P lies opposite sides of $x - 4y = 0$)

$$\Rightarrow 0 < a \text{ and } a \in (-\infty, 0) \cup (2, \infty)$$

$$\Rightarrow a \in (2, 4)$$

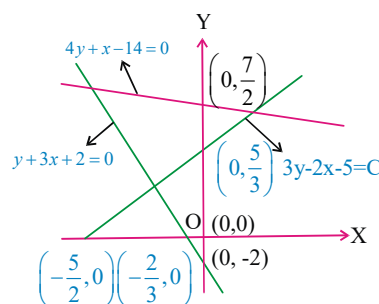
19. $D < 0 \Rightarrow b^2 - 4 < 0 \Rightarrow -2 < b < 2$ but $b > 0$

$$\therefore 0 < b < 2; \text{ i.e., } b \in (0, 2)$$

20. Draw the diagram

21. Clearly point of intersection of L_1 and L_2 , is $(0, k)$ which lies on y - axis.

- 22.



from the figure, it is clear that $\frac{5}{3} \leq \alpha \leq \frac{7}{2}$

23. Point of intersection = $\left(\frac{1+|a|}{1+a}, \frac{a|a|-1}{a+1} \right)$

since $x > 0, y > 0$ we have $a+1 > 0$ and $a|a| > 1$

as $|a| \geq 0, a|a| > 1$ we get $a > 0$ thus $a^2 > 1$

or $a > 1$

24. Point $(k, -k)$ satisfies both lines then

$$c = -2ak, d = -3bk \Rightarrow \frac{c}{-2a} = \frac{d}{-3b}$$

25. $\begin{vmatrix} 1 & a & a \\ b & 1 & b \\ c & c & 1 \end{vmatrix} = 0$; Use det properties

26. $(2a + 3b)^2 - c^2 = 0$

27. $a^2 - (b + c)^2 = 0$

$$(a - b - c)(a + b + c) = 0$$

$$\text{Either } a - b - c = 0 \text{ or } a + b + c = 0$$

$$\Rightarrow -a + b + c = 0$$

$\Rightarrow$ Family of lines passing through $(-1, 1)$ and $(1, 1)$.

28. t_1, t_2, t_3 roots of $y + xt = 2at + at^3$

29. The family of lines

$$(x + y - 1) + \lambda(2x + 3y - 5) = 0 \text{ passes}$$

through a point such that

$$x + y - 1 = 0, \quad 2x + 3y - 5 = 0$$

30. a, b, c are in A.P.
 31. Let slope of another side is m
 $\tan 60^\circ = \left| \frac{m-1}{1+m} \right|$; $m = -2 \pm \sqrt{3}$
 32. ΔPQR is right angle triangle
 radius = altitude
 33. Consider $B(-p, 0) C(p, 0) A(0, q)$
 34. Since m_1, m_2 are the roots of the
 $x^2 - ax - a - 1 = 0$, so

$$m_1 + m_2 = a; m_1 m_2 = -(a+1)$$

$$\Rightarrow (m_1 - m_2)^2 = (m_1 + m_2)^2 + 4m_1 m_2$$

$$= a^2 + 4(a+1) = (a+2)^2$$

so the required area is

$$\Delta = \pm \frac{a^2 (a+2)}{2(a+1)}$$

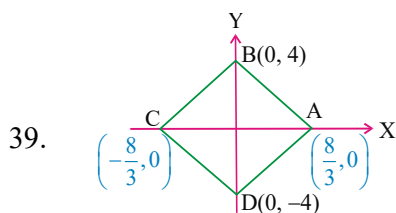
since area is positive quantity, so we get two answers

35. Find the area of the triangle formed by the lines
 $x + y = 2$, $2x - 2y = 1$, $x = 0$
 36. Let D is the mid point of AB

$$OD = \frac{1}{\sqrt{2}}, AD = \sqrt{\frac{71}{2}}, AB = 2AD$$

37. required area = $\frac{p^2}{\sqrt{3}}$ where p is distance between
 parallel lines

38. Area of $\Delta^{le} = \frac{\Delta^2}{2\lambda_1 \lambda_2 \lambda_3}$ where Δ is
 determinant of 3 lines



Enclosed Area ABCD = 4 Area AOB

$$= 4 \times \frac{1}{2} \times \frac{8}{3} \times 4 = \frac{64}{3} \text{ sq. units.}; \therefore f(n) = 2^n$$

$$\text{required area} = \frac{1}{3} f(6)$$

$$40. \frac{1}{4} \int_0^2 (x+k) dx = \frac{1}{2} (6).$$

41. From diagram $3\theta = 45^\circ \Rightarrow \theta = 15^\circ$
 42. from diagram $3\theta = 180^\circ \Rightarrow \theta = 60^\circ$
 43. Intersecting $2x + 3y - 5 = 0$ and
 $3x - 5y + 2 = 0$ is $(1, 1)$
 Find the image of $(1, 1)$ w.r.to $x + y = \sqrt{2}$
 44. If we reflect $y = |x - 2|$ in y-axis, it will
 becomes $y = |-x - 2| = |x + 2|$. The reflected
 lines are $y = x + 2$, $y = -x - 2$. Their combined
 equation is $(y - x - 2)(y + x + 2) = 0$
 $\Rightarrow y^2 - (x + 2)^2 = 0$
 $\Rightarrow y^2 - x^2 - 4x - 4 = 0$

45. using symmetric form we get $A = (1, \sqrt{3})$, G
 divides $\overline{AD}$ in the ratio 2:1

46. $AC : 3x - 2y = k$
 $-3 - 6 = k$, $3x - 2y = -9$
 $M : (1, 6)$, $\therefore C(3, 9)$ $G(2, 4)$

47. Triangle is acute angled
 48. Triangle is obtuse angled

49. Let $A(h, k)$, $D(\alpha, \beta)$ be the point on BC
 Then

$$\left(\frac{2\alpha + h}{3}, \frac{2\beta + k}{3} \right) = (0, 0) \text{ \& } \alpha + \beta - 2 = 0,$$

$$\text{and } \left(\frac{k-0}{h-0} \right) (-1) = -1$$

gives $\alpha = \beta = 1$ and $h = k = -2$
 Hence $A(-2, -2)$

50. $m_1 = \frac{1}{2}$, $m = \frac{-2}{3}$, m_2 is slope of reflected line

$$\text{use } \frac{m_1 - m}{1 + mm_1} = \frac{m - m_2}{1 + mm_2}$$

51. $ax + by = 1$ will be one of the bisectors of the given
 line. Equation of bisectors of the given lines are

$$\frac{3x + 4y - 5}{5} = \pm \left(\frac{5x - 12y - 10}{13} \right)$$

$$\Rightarrow 64x - 8y = 115 \text{ or } 14x + 112y = 15$$

$$\Rightarrow a = \frac{64}{115}, b = \frac{8}{115} \text{ or } a = \frac{14}{15}, b = \frac{12}{115}$$

$$52. x_1 = -10, y_1 = 2 \text{ \& } x_2 = \frac{-19}{2}, y_2 = \frac{3}{2}$$

$$x_3 = \frac{-28}{3}, y_3 = \frac{4}{3}$$

53. Equation of AB is $x + y = 21$
Number of integral solutions of $x + y < 21$
is $20C_2$

54. Verification

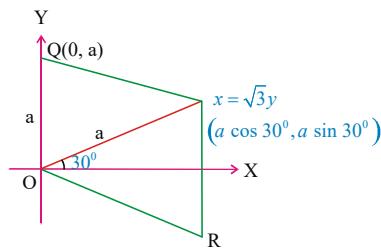
55. Verification

56. verification

57. Clearly Q, R are the two positions of the third vertex.

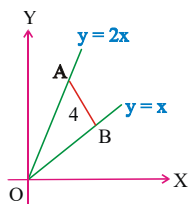
$$P(a \cos 30^\circ, a \sin 30^\circ), Q(0, a), R(a \cos 30^\circ, -a \sin 30^\circ)$$

$$\text{i.e. } P\left(\frac{\sqrt{3}a}{2}, \frac{a}{2}\right), Q(0, a), R\left(\frac{\sqrt{3}a}{2}, -\frac{a}{2}\right)$$



58. Let $B = (\alpha, \alpha)$ and middle point AB is (h, k)

$$\text{Then, } A \equiv (2h - \alpha, 2k - \alpha)$$



lies on $y = 2x$ then $\alpha = 4h - 2k \dots (1) \quad |AB| = 4$

$$\Rightarrow [h - (4h - 2k)]^2 + [k - (4h - 2k)]^2 = 4$$

$$\Rightarrow (-3h + 2k)^2 + (-4h + 3k)^2 = 4$$

$$\text{or } 25h^2 + 13k^2 - 36hk = 4$$

$$\text{Required locus is } 25x^2 + 13y^2 - 36xy - 4 = 0$$

$$\text{Here, } h^2 < ab \text{ and } \Delta \neq 0$$

$\therefore$ ellipse



LEVEL - IV

1. (A) : If $(a_1x + b_1y + c_1) + (a_2x + b_2y + c_2)$

$+ (a_3x + b_3y + c_3) = 0$, then the lines

$$a_1x + b_1y + c_1 = 0,$$

$$a_2x + b_2y + c_2 = 0, a_3x + b_3y + c_3 = 0$$

can not be parallel

(R): If sum of three straight lines is identically 0 then they are either concurrent or parallel

1) A and R are true and R is the correct explanation of A

2) A and R are true and R not is the correct explanation of A

3) A is true R is False 4) A is False R is True

2. A: $(3, 2)$ is lies above the line $x + y + 1 = 0$

R: If the point $P(x_1, y_1)$ lies above the line

$$L = ax + by + c \text{ then } \frac{L(x_1, y_1)}{b} > 0$$

1) A and R are true and R is the correct explanation of A

2) A and R are true and R not is the correct explanation of A

3) A is true R is False 4) A is False R is True

3. Assertion (A): If the angle between the lines $kx - y + 6 = 0, 3x + 5y + 7 = 0$ is $\pi/4$ one value of k is -4

Reason (R): If θ is angle between the lines

$$\text{with slopes } m_1, m_2 \text{ then } \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|.$$

1) A and R are true and R is the correct explanation of A

2) A and R are true and R not is the correct explanation of A

3) A is true R is False 4) A is False R is True

4. I : Every first degree equation in x and y is $ax + by + c = 0, |a| + |b| \neq 0$ represent a straight line

II : Every first degree equation in x and y can be convert into slope intercept form

Then which of the following is true

1) Only I

2) only II

3) both I & II

4) neither I nor II

5. I : Length of the perpendicular from (x_1, y_1) to

the line $ax+by+c=0$ is $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

II : The equation of the line passing through $(0,0)$ and perpendicular to $ax+by+c=0$ is $bx-ay=0$

Then which of the following is true.

- 1) only I 2) only II
3) both I & II 4) neither I nor II
6. I : The ratio in which $L \equiv ax+by+c=0$ divides the line segment joining $A(x_1, y_1)$ $B(x_2, y_2)$ is

$$\frac{-L_{11}}{L_{22}}$$

II: the equation of the line in which (x_1, y_1) divides the line segment between the coordinate axes in the ratio $m:n$ is

$$\frac{nx}{x_1} + \frac{my}{y_1} = m + n$$

Then which of the following is true

- 1) only I 2) only II
3) both I & II 4) neither I nor II
7. I: A straight line is such that the algebraic sum of the distance from any no. of fixed points is zero. Then that line always passes through a fixed point

II: The base of the triangle lie along the line $x=a$ and is of length a . If the area of the triangle is a^2 then the third vertex lies on $x=-a$ or $x=3a$.

Then which of the following is true.

- 1) only I 2) only II
3) both I & II 4) neither I nor II
8. Statement I: Normal form of line $x + y = \sqrt{2}$ is

$$x \cos \frac{\pi}{4} + y \sin \frac{\pi}{4} = 1$$

Statement II: The ratio in which the perpendicular through $(4,1)$ divides the line joining $(2,-1)$, $(6,5)$ is 5:8

Which of the above statement (s) is/are true

- 1) Only I 2) Only II
3) Both I and II 4) Neither I nor II

9. The lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersect L_3 at R

Statement - 1 : The ratio PR : RQ equals $2\sqrt{2} : \sqrt{5}$.

Statement - 2 : In any triangle, bisector of an angle divides the triangle into two similar triangles. [AIEEE - 2011]

- (1) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not a correct explanation for Statement - 1
(2) Statement - 1 is true, Statement - 2 is false.
(3) Statement - 1 is false, Statement - 2 is true.
(4) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1

10. Observe the following list with respect to the line $ax+by+c=0$

List I

List II

A) Perpendicular distance from $(0,0)$ 1) $-\frac{c}{b}$

B) X-intercept of the line 2) $\left(\frac{-c}{a}, \frac{-c}{b}\right)$

C) Y-intercept of the line 3) $\frac{|c|}{\sqrt{a^2 + b^2}}$

D) Circumcentre of triangle OAB 4) $\left[\frac{-c}{2a}, \frac{-c}{2b}\right]$

where A, B are X and Y intercepts 5) $-\frac{c}{a}$

Then the correct answer is

- | | | | | | | | |
|------|---|---|---|------|---|---|---|
| A | B | C | D | A | B | C | D |
| 1) 3 | 5 | 1 | 2 | 2) 3 | 5 | 1 | 4 |
| 3) 3 | 4 | 1 | 5 | 4) 1 | 2 | 3 | 4 |

11. observe the following
- | | |
|--|------------------|
| column I | column II |
| A) the area bounded by the curve $\max\{ x , y \} = 1$ is | P) 3 |
| B) if the point (a,a) lies between the lines $ x+y =3$ then number of values of $[a]$ is (where $[.]$ denotes the greatest integer function) | Q) 2 |
| C) Number of integral values of b for which the origin and the point(1,1) lie on the same side of the st.line | R) 4 |
- $a^2x + aby + 1 = 0$ for all $a \in \mathbb{R} \sim \{0\}$
- 1) $A \rightarrow R, B \rightarrow Q, C \rightarrow P$
 - 2) $A \rightarrow R, B \rightarrow P, C \rightarrow Q$
 - 3) $A \rightarrow P, B \rightarrow Q, C \rightarrow R$
 - 4) $A \rightarrow P, B \rightarrow R, C \rightarrow Q$
12. Equation of line passing through (1,3), perpendicular to $2x-3y+4=0$ is $ax+by+c=0$ ($a>0$) then ascending order of a, b, c is
- 1) a, c, b
 - 2) c, b, a
 - 3) c, a, b
 - 4) a, b, c

LEVEL-IV - KEY

- | | | | | | |
|------|------|------|-------|-------|-------|
| 1) 4 | 2) 1 | 3) 1 | 4) 1 | 5) 3 | 6) 3 |
| 7) 3 | 8) 3 | 9) 2 | 10) 2 | 11) 1 | 12) 2 |

LEVEL-IV - HINTS

1. Given lines may be parallel.
2. R is the correct explanation of A.
3. Using angle formula between the lines
4. Standard result
5. Standard result
6. Standard result
7. I: Consider the points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
Algebraic sum of the perpendicular distance = to $\frac{a(x_1 + x_2 + \dots + x_n) + b(y_1 + y_2 + \dots + y_n) + nc}{\sqrt{a^2 + b^2}}$
II: Consider the points on the line $x = a$ are A(a,0) B(a,a)
P(x,y) area of the triangle = a^2

8. I) normal form line $x \cos \alpha + y \sin \alpha = p$
II) $-L_{11} : L_{22}$
9. $P(2, -2), Q(1, -2)$
Equation of angular bisector OR is $(\sqrt{5} + 2\sqrt{2})x = (\sqrt{5} - \sqrt{2})y$
 $\therefore PR : RQ = 2\sqrt{2} : \sqrt{5}$
10. A) the perpendicular distance from (0,0) to $ax + by + c = 0$ is $\frac{|c|}{\sqrt{a^2 + b^2}}$
B) X-Intercept $\frac{-c}{a}$
C) Y-intercept $\frac{-c}{b}$
D) In right angle triangle circumcenter is mid point of hypotenuse is
11. (A) $\therefore \max\{|x|, |y|\} = 1$
If $|x| = 1$ and if $|y| = 1$,
then $x = \pm 1$ and $y = \pm 1$
 $\therefore$ Required area = 2×2
(B) the line $y = x$ cuts the lines $|x + y| = 3$ ie,
 $x + y = \pm 3$
at $x = \pm 3/2, y = \pm 3/2$ or $(-3/2, -3/2)$ and $(3/2, 3/2)$
then $-3/2 < a < 3/2, \therefore 0 \leq |a| < 3/2$
 $\therefore [a] = 0, 1$
(C) since (0, 0) and (1, 1) lie on the same side.
So, $a^2 + ab + 1 > 0$
 $\therefore$ Coefficient of a^2 is > 0
 $\therefore D < 0$
 $b^2 - 4 < 0$
or $-2 < b < 2$
 $\Rightarrow b = -1, 0, 1$
 $\therefore$ Number of values of b is 3.
12. Line equation is $3x + 2y - 9 = 0$.